HUMBOLDT-UNIVERSITÄT ZU BERLIN

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Development and Evaluation of an R package for Nutrient Solution Formulation in Horticulture

Dilân Olmo Şakar

Program: M.Sc. International Horticultural Science

Institute: Thaer-Institute of Agricultural and Horticultural Sciences

Department: Plant Nutrition and Fertilization

Supervisors: Prof. Dr. Eckhart George, Dipl.-Ing. Dr. Olivier Duboc

Institutions: Humboldt University of Berlin (HU) & University of Natural Resources and Life Sciences, Vienna (BOKU)

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Abstract

Efficient nutrient-solution formulation is fundamental to controlled horticultural production, yet in practice it is often based on fixed recipes or iterative manual adjustment. This thesis addresses that limitation through the development of NutriCalc, a modular open-source R framework for reproducible nutrient-solution design.

The central objective was to formalise nutrient formulation as a mathematically defined problem under horticultural and chemical constraints. The formulation task was expressed as a constrained linear optimisation problem and solved using non-negative least squares, ensuring non-negative fertilizer quantities. The framework combines stoichiometric conversion, unit transformation, charge-balance-based evaluation and estimation of pH and electrical conductivity, thereby linking nutrient targets to key physicochemical properties of the final solution.

NutriCalc was implemented as both an R package for programmable workflows and a Shiny-based graphical user interface for interactive use. Overall, the thesis demonstrates that nutrient-solution design can be shifted from recipe-driven practice to a reproducible, optimisation-based workflow grounded in solution chemistry. NutriCalc provides a transparent computational foundation for horticultural research, teaching and applied nutrient management.

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List of Abbreviations

A-BAK Stock solution A [A-Stammlösung / stock tank A]

B-BAK Stock solution B [B-Stammlösung / stock tank B]

BK Base Capacity [Basenkapazität]

BOKU University of Natural Resources and Life Sciences, Vienna [Universität für Bodenkultur Wien]

BWB Berliner Wasserbetriebe

CO_2(aq) Aqueous carbon dioxide

CAN Calcium ammonium nitrate

CRC CRC Handbook of Chemistry and Physics

CSV Comma-Separated Values

DWC Deep Water Culture

EC Electrical Conductivity

EDTA Ethylenediaminetetraacetic acid

Fe-EDTA Iron ethylenediaminetetraacetate

GUI Graphical User Interface

HU Humboldt-Universität zu Berlin

ME Mean Error

MAE Mean Absolute Error

meq L^{-1} Milliequivalents per liter

mmol L^{-1} Millimoles per liter

N_A Avogadro constant

NNLS Non-Negative Least Squares

pH Negative logarithm of the hydrogen ion activity

PHREEQC PH REdox EQuilibrium; Geochemical speciation and reaction-modeling software developed by the U.S. Geological Survey

R The R programming language and software environment for statistical computing

RDW Reuse drainage water

SI International System of Units [Système international d'unités]

SK Acid Capacity [Säurekapazität]

SSE Sum of Squared Errors

SSRE Sum of Squared Relative Errors

TCE Total Charge Equivalents

WUE Water Use Efficiency

1 Introduction

Precise nutrient solution formulation is fundamental to controlled horticultural production, particularly in hydroponic and other soilless cultivation systems. In these systems, plant nutrition depends directly on the chemical composition of the supplied solution. Inaccuracies in formulation can therefore affect plant performance, experimental validity and resource-use efficiency.

In practice, however, nutrient solution design remains a complex task. Fertilizer salts often supply several nutrients simultaneously, local irrigation water contributes additional ions and target concentrations must be translated between different chemical and commercial representations. As a result, manual formulation and opaque spreadsheet-based workflows are prone to error, especially when transparency, reproducibility and extensibility are required.

These challenges are particularly relevant in horticultural research, where treatments are often designed to manipulate individual nutrients while keeping other factors as constant as possible. Because ions are chemically coupled through shared salts, electroneutrality and solution chemistry, such designs require computationally explicit methods rather than ad hoc adjustment. Despite this need, no dedicated modular open-source framework in R is currently available for transparent nutrient solution formulation and analysis in horticultural applications.

This thesis addresses that gap through the development of NutriCalc, an open-source computational framework in R for the formulation, analysis and optimisation of nutrient solutions. The work formalises nutrient formulation as a constrained linear optimisation problem solved using non-negative least squares and extends this workflow through stoichiometric conversion, charge-balance-based evaluation, source-water correction and estimation of pH and electrical conductivity. In addition to the computational backend, the framework is implemented as a Shiny-based graphical user interface to support practical use by researchers and horticultural practitioners. The overarching objective of the thesis is to provide a reproducible, chemically coherent and extensible framework that shifts nutrient solution formulation from a recipe-based practice to a transparent computational workflow.

1.1 Need for transparent and reproducible formulation tools

A reconstruction of nutrient calculations reported in the literature illustrates the importance of explicit conversion pipelines. For example, J. J. Jones et al. (2024) describe a control nutrient solution for CUBES Circle based on defined molar concentrations of fertilizer salts: 5.15 mM $Ca(NO_3)_2$, 8.79 mM KNO_3, 2.36 mM $MgSO_4$ and 2.94 mM KH_2PO_4. The solution was subsequently scaled to 10 % strength. Recalculating selected nutrient concentrations shows that the final values depend strongly on how stoichiometric coefficients, dilution factors and ion attribution are applied.

Using elemental molar masses, stoichiometric attribution and the stated 10 % dilution, the expected concentrations for Ca, Mg and nitrate are:

$$c(\mathrm{Ca}) = 5.15\,\mathrm{mmol\,L^{-1}} \times 40.08\,\mathrm{mg\,mmol^{-1}} \times 0.1 = 20.64\,\mathrm{mg\,L^{-1}}.$$

$$c(\mathrm{Mg}) = 2.36\,\mathrm{mmol\,L^{-1}} \times 24.31\,\mathrm{mg\,mmol^{-1}} \times 0.1 = 5.74\,\mathrm{mg\,L^{-1}}.$$

$$c(\mathrm{NO_3}^-) = (2 \times 5.15 + 8.79)\,\mathrm{mmol\,L^{-1}} \times 62.01\,\mathrm{mg\,mmol^{-1}} \times 0.1 = 118.38\,\mathrm{mg\,L^{-1}}.$$

These reconstructed values differ from the published values for Ca, Mg and NO_3^-. The reported Ca concentration of 82.05 mg $\mathrm{L^{-1}}$ is consistent with using the molar mass of $Ca(NO_3)_2$ and an effective concentration of 5.0 mmol $\mathrm{L^{-1}}$, rather than calculating elemental Ca:

$$5.0\,\mathrm{mmol\,L^{-1}} \times 164.09\,\mathrm{mg\,mmol^{-1}} \times 0.1 = 82.05\,\mathrm{mg\,L^{-1}}.$$

The Mg concentration of 57.37 mg $\mathrm{L^{-1}}$ is consistent with omission of the 10 % dilution factor, whereas the nitrate concentration of 63.87 mg $\mathrm{L^{-1}}$ is consistent with attributing nitrate only to $Ca(NO_3)_2$ and omitting the contribution from KNO_3:

$$2.36\,\mathrm{mmol\,L^{-1}} \times 24.31\,\mathrm{mg\,mmol^{-1}} = 57.37\,\mathrm{mg\,L^{-1}}.$$

$$(2 \times 5.15)\,\mathrm{mmol\,L^{-1}} \times 62.01\,\mathrm{mg\,mmol^{-1}} \times 0.1 = 63.87\,\mathrm{mg\,L^{-1}}.$$

These discrepancies do not necessarily affect the overall conclusions of the study. However, they demonstrate how small ambiguities in conversion logic can lead to substantial differences in reported nutrient concentrations. The example therefore underlines the need for calculation frameworks in which unit conversions, stoichiometric coefficients, dilution factors and ion contributions are encoded explicitly and reproducibly. This is particularly relevant for integrated production systems, where nutrient streams may be reused across modules and quantitative consistency is required.

1.2 Background and motivation

Precise nutrient solution formulation involves selecting and combining soluble fertilizer salts to deliver specific concentrations of essential elements in desired chemical forms and ratios (J. B. Jones, 2014). While standard formulations were published since the mid-19th century (Hoagland, 1920; Hoagland & Arnon, 1938; Knop, 1860; Shive, 1915; Stout & Arnon, 1939) and were meticulously collected by Hewitt (1966), practitioners frequently need to tailor them to local water quality, crop-specific requirements or available fertilizers, creating a complex and often error-prone challenge (Resh, 2013; Sonneveld, 2009).

In plant nutrition experiments, treatments are typically designed to vary individual nutrients while holding others constant (ceteris paribus) to investigate the effect of a single nutrient element. However, due to the interdependencies among ions and their shared presence in various fertilizer salts, such isolation is rarely achievable in practice. Researchers must therefore account for the introduction of accompanying ions, which may affect plant responses and experimental validity.

Creating accurate nutrient solutions under such constraints is time-consuming and sometimes mathematically infeasible, underscoring the need for computational tools that enable efficient, accurate and reproducible formulation under realistic chemical and experimental conditions.

The nutrient formulation problem is inherently mathematical. Many fertilizer salts contribute multiple nutrients (e.g., $Ca(NO_3)_2$ supplies both calcium and nitrate), leading to an interdependent system. This can be modelled as a system of linear equations:

$$A \cdot x = b$$

where:

- A is the nutrient matrix, with each row representing a nutrient and each column a fertilizer salt;
- x is the vector of unknown fertilizer quantities;
- b is the vector of target nutrient concentrations.

Due to overlapping contributions and limited degrees of freedom, exact solutions may not exist. Thus, approximate solutions are needed, with the additional constraint that all fertilizer quantities must be non-negative. This motivates the use of non-negative least squares (NNLS) optimisation, which minimises the squared error between the achieved and target nutrient vectors under the constraint $x \geq 0$.

Moreover, real-world use requires support for various chemical notations such as conversions between elemental and oxide forms (e.g., $P \leftrightarrow P_2O_5$, $S \leftrightarrow SO_4^{2-}$) and integration with commercially available fertilizers labelled in % w/w or expressed in units such as mmol L^{-1} or mg L^{-1}. Despite the existence of some basic Windows applications and Excel tools, there is currently no open-source, modular and transparent software solution in R that offers this level of flexibility and chemical accuracy for horticultural applications.

1.3 Objectives and research questions

The main objective of this thesis was to develop and evaluate a computational framework in R for the formulation, analysis and optimisation of nutrient solutions in controlled-environment horticulture. The framework was intended to support accurate and efficient nutrient planning while improving reproducibility and transparency in both research and practical applications. More specifically, the thesis sought to establish tools for constructing nutrient matrices from molecular fertilizer formulas and for converting fertilizer compounds into their elemental equivalents by means of stoichiometric calculations. On this basis, nutrient formulation was formalised as a non-negative linear optimisation problem and solved using Non-Negative Least Squares (Mullen & van Stokkum, 2024).

Beyond nutrient matching, the framework was designed to estimate key physicochemical properties of the final solution, in particular pH and electrical conductivity, from ion concentrations and molar conductivity data (McCleskey et al., 2012). In addition, the workflow was designed to make the distribution of nutrients across fertilizer sources transparent. To ensure practical accessibility, the computational workflow was implemented both as an R package and as a graphical user interface using the Shiny framework, enabling interactive use by non-technical users (Chang et al., 2025).

The work was guided by the following research questions:

1. Can nutrient solution formulation be represented as a reproducible stoichiometric optimisation problem using chemically defined fertilizer compounds?
2. To what extent can NNLS-based formulation reconstruct published horticultural nutrient solution recipes?
3. How closely do the implemented pH and EC estimation routines reproduce selected reference systems for phosphate buffering and KCl conductivity?
4. How does the NutriCalc workflow compare with a published formulation approach by Sonneveld (2009)?
5. Can the framework handle practical formulation constraints, including restricted fertilizer availability, source-water correction, acid adjustment and mixed-water formulation?

Taken together, this thesis makes three contributions. First, it formalises nutrient solution formulation as a constrained stoichiometric optimisation problem under non-negativity constraints. Second, it implements this formulation in a reproducible R-based workflow that links molecular fertilizer formulas, nutrient targets, source-water composition and physicochemical output variables. Third, it evaluates the framework through recipe reconstruction, reference-system checks for pH and EC, comparison with a published formulation approach and applied case studies. The contribution is therefore computational-methodological rather than experimental: the thesis evaluates whether nutrient solution formulation can be made transparent, reproducible and chemically coherent within the implemented formulation domain.

2 Theoretical foundations and existing approaches

2.1 Principles of nutrient solution formulation and experimental design in plant nutrition

This section develops a formal framework, expressed in milliequivalent (meq) notation, that derives a classification of nutrient solution experimental designs from first principles and clarifies the interpretational limits of nutrient experimentation. It addresses horticultural plant nutrition experiments aimed at evaluating nutrient composition and its effects under chemically and statistically controlled conditions. It does not address other types of biological experiments, such as studies on uptake mechanisms, isotope tracing, signaling pathways, or related physiological processes.

Experiments in plant nutrition seek to identify causal relationships between specific nutrients and plant responses such as growth, yield, vitality and physiological performance. Despite the apparent simplicity of manipulating nutrient concentrations, such experiments are fundamentally constrained by chemical laws that limit causal identifiability. In particular, electroneutrality (Kalka, 2024) and the statistical requirement of ceteris paribus jointly restrict how nutrient solutions can be manipulated and how resulting effects may be interpreted. The main difficulty associated with nutrient solution formulation is that any attempt to add an ion to a nutrient solution necessarily requires the accompanying supply of an ion of opposite charge (Savvas & Adamidis, 1999). In addition to charge balance, many nutritionally relevant components participate in acid–base equilibria and therefore exist as families of chemical species (Appelo, 2005; Haynes, 2017).

A nutrient solution may be represented as a vector of its ionic charge equivalents:

$$\mathbf{S}_{\text{meq}} = \left(\text{K}^{+},\ \text{Ca}^{2+},\ \text{Mg}^{2+},\ \dots\ \middle|\ \text{NO}_3^{-},\ \text{Cl}^{-},\ \text{SO}_4^{2-},\ \dots\right)$$

where each component is expressed in milliequivalents per liter (meq L^{-1}). For any ion i,

$$\text{meq}_i = |z_i|\, c_i,$$

with z_i denoting ionic charge and c_i the molar concentration. Expressing solution composition in meq makes ionic charge balance explicit and avoids ambiguity associated with heterovalent ions.

Not all vectors in this space are chemically admissible. All aqueous nutrient solutions must satisfy electroneutrality, which in meq notation takes the form

$$\sum \text{meq}_{\text{cat}} - \sum \text{meq}_{\text{an}} = 0.$$

The sums include all charged species present, including H^+ (formally H_3O^+) and OH^-. Electroneutrality implies that no ionic component can be altered independently of ions carrying opposite charge. Any experimental manipulation therefore induces compensatory changes elsewhere in the ionic system. Activity coefficients, complexation and ion pairing may alter speciation but do not remove the electroneutrality constraint.

Many nutritionally relevant components participate in acid–base equilibria and therefore exist as families of chemical species. For example, phosphate occurs as $H_2PO_4^-$, HPO_4^{2-} and PO_4^{3-}, while carbonate occurs as CO_2, HCO_3^- and CO_3^{2-}. Let c_i^{tot} denote the total analytical concentration of component i. The total charge contribution of this component to the solution can be written as

$$\text{meq}_i^{\text{tot}} = \sum_j |z_{ij}|\, c_{ij},$$

where c_{ij} is the concentration of species j belonging to component i. Changes in pH redistribute charge among species with different valencies as shown for phosphate in Figure 2.1, thereby altering the effective ionic composition of the solution even when total analytical concentrations are held constant. Although pH can be experimentally controlled using buffers or acid–base additions, such control necessarily introduces additional ionic components and therefore does not decouple pH from nutrient composition in a causal sense.

From a causal-inference perspective, identification of an effect requires isolating variation in a target factor while holding other causally relevant factors fixed or blocked (ceteris paribus). In plant nutrition experiments, strict independent variation of a single ionic species is generally not possible, because electroneutrality and acid–base equilibria couple ionic charge, pH, ionic strength and osmotic potential. As a result, strict ceteris paribus conditions for individual ions cannot be achieved in bulk nutrient solutions; they can only be approximated within constrained experimental frameworks. Within this framework, classical experimental designs in plant nutrition can be derived as three distinct classes of experiments, defined by the invariants they preserve in meq-space.

Here, Δ denotes the change induced by the experimental manipulation, i.e. $\Delta x := x_{\text{treatment}} - x_{\text{control}}$ (or, more generally, the difference between two solution states). The first class consists of qualitative, or ion-substitution, experiments. In these experiments, one ionic species is replaced by another while total cation and total anion charge equivalents are kept constant:

$$\Delta\text{meq}_i + \Delta\text{meq}_j = 0.$$

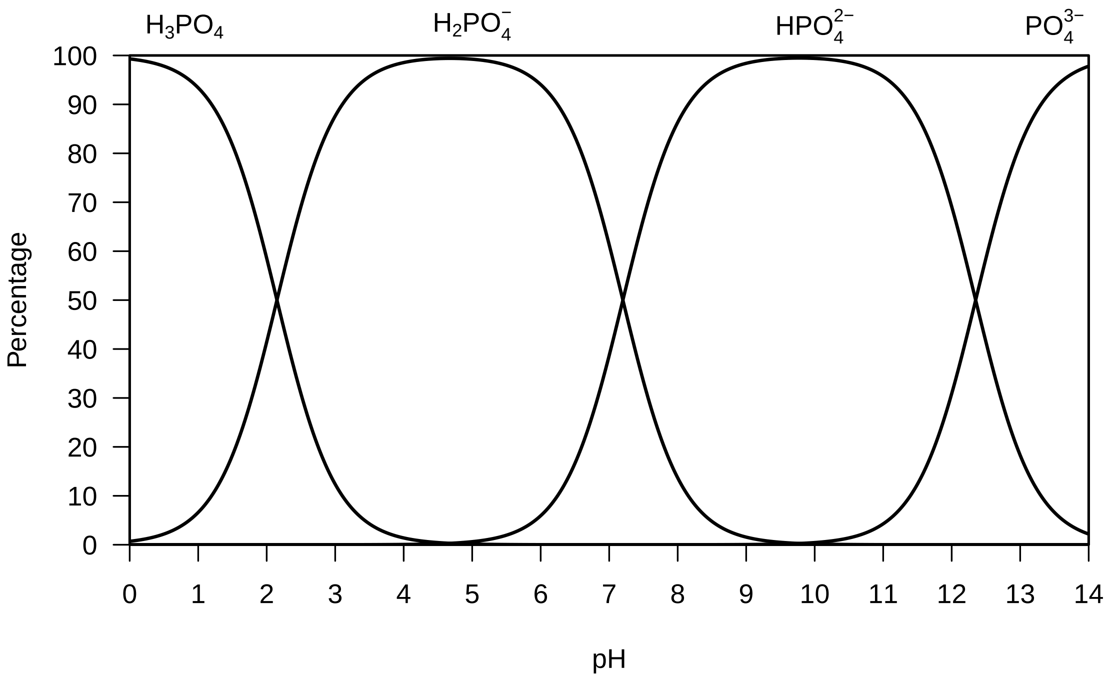


Figure 2.1: Phosphate speciation (percent) as a function of pH, calculated from phosphate acid dissociation constants (Haynes, 2017).

When the substituted ions have the same charge, the substitution is isovalent (e.g. $K^+ \leftrightarrow Na^+$ or $Cl^- \leftrightarrow NO_3^-$). When the charges differ, the substitution is heterovalent, for example replacing 2 meq of K^+ (2 mmol) with 2 meq of Ca^{2+} (1 mmol). Qualitative substitution experiments provide the highest degree of causal specificity achievable within whole-solution manipulations at fixed charge totals, although their interpretation remains conditional on buffering and counter-ion choice.

The second class comprises quantitative, or salt addition (or removal), experiments. Here, the total amount of dissolved salt is changed, which necessarily alters both total cation and total anion charge equivalents by the same non-zero amount:

$$\Delta \sum \text{meq}_{\text{cat}} = \Delta \sum \text{meq}_{\text{an}} \neq 0.$$

Electroneutrality is preserved automatically, but ionic strength, electrical conductivity and osmotic potential change and pH may shift depending on buffering. Because at least two ionic components vary simultaneously, plant responses in such experiments reflect composite effects of nutrient ions, counter-ions and general salinity.

The third class treats quantity as an emergent quality through solution-strength series. In this case, all charge equivalents in the solution are scaled by a constant factor k:

$$\mathbf{S}_{\text{meq},k} = k\,\mathbf{S}_{\text{meq},1}.$$

This transformation preserves ion ratios expressed in meq while increasing or decreasing total charge density, ionic strength and osmotic potential. Although composition remains constant at the level of charge ratios, pH and ionic speciation may shift because acid–base equilibria are nonlinear. As a result, solution strength functions as an environmental condition rather than a simple quantitative variable.

Across these three experiment types, causal specificity for individual ion identity typically decreases as the number of inseparable or uncontrolled variables increases. Qualitative substitution experiments offer the highest degree of causal specificity, whereas quantitative salt manipulations and solution-strength series progressively trade causal resolution for ecological or horticultural relevance. These experiment classes are therefore best understood as distinct projections of a chemically constrained ionic system, most transparently described in milliequivalent space. Based on the invariants preserved in meq-space, the resulting classification of nutrient solution experimental designs is summarized in Table 2.1. Qualitative substitutions preserve total charge equivalents, whereas salt additions and solution-strength series necessarily alter total charge density, EC and osmotic potential.

Table 2.1: Classification of nutrient solution experimental designs in milliequivalent space.

Experiment class	Core manipulation	Charge-balance condition	Example
I.a Qualitative (isovalent)	Equal-meq substitution within the same charge group	$\Delta \mathrm{meq}_i + \Delta \mathrm{meq}_j = 0$	$K^+ \leftrightarrow Na^+$
I.b Qualitative (heterovalent)	Equal-meq substitution between ions of different charge	$\Delta \mathrm{meq}_i + \Delta \mathrm{meq}_j = 0$	2 mmol $K^+ \leftrightarrow$ 1 mmol Ca^{2+}
II Quantitative	Salt addition or removal	$\Delta \sum \mathrm{meq}_{\mathrm{cat}} = \Delta \sum \mathrm{meq}_{\mathrm{an}} \neq 0$	$\uparrow$ KCl
III Quantitative-as-qualitative	Scaling of the whole solution by factor k	$\mathbf{S}_{\mathrm{meq},k} = k\,\mathbf{S}_{\mathrm{meq},1}$	¼×, ½×, 1×

A comparison between nitrate and ammonium nutrition illustrates the limits of simple ion-substitution experiments. Because NO_3^- and NH_4^+ carry opposite charge, direct replacement of nitrate by ammonium is not electroneutral and requires compensatory changes in accompanying ions. Such treatments are therefore not pure single-ion substitutions, but compound treatment contrasts whose interpretation depends on the counter-ion strategy used to preserve charge balance.

From a plant nutrition perspective, comparisons between nutrient solutions of different provenance are only informative to the extent that differences in ionic composition are maintained or eliminated in a controlled manner. If two solutions of different origin, such as wastewater-derived (e.g. animal or human urine) and mineral solutions, are formulated to the same ionic composition, then the specifically nutritional dimension of the comparison is largely removed, reducing the question to a Ship of Theseus problem of identity. Under such conditions, any remaining differences are more appropriately interpreted in terms of economic factors associated with source, production, transport, cost and sustainability rather than nutrient composition itself.

2.2 Stoichiometry and salt–ion relationships

In hydroponics, nutrients are supplied by dissolving fertilizer salts in water to prepare a nutrient solution (Resh, 2013). The formulation of such solutions is complicated by the declaration conventions used for commercial fertilizers. Nutrient contents are commonly expressed in oxide form rather than as elemental concentrations or ionic species. Thus, potassium is typically declared as K_2O rather than as K or ionic K^+, while phosphorus is conventionally declared as P_2O_5 rather than as P or as the pH-dependent phosphate species $H_2PO_4^-$, HPO_4^{2-}, or PO_4^{3-}. Although these conventions are historically established, they introduce an unnecessary level of abstraction and increase the susceptibility of nutrient calculations to error.

A more serious limitation is that fertilizer labels do not necessarily disclose all chemically relevant constituents. Consequently, attempts to reconstruct complete fertilizer compositions in ionic form may fail to account for the complete ionic composition. This can occur, for example, when ions such as chloride are present but are not part of the mandatory nutrient-declaration structure. Under Regulation (EU) 2019/1009, nutrient declaration requirements depend on the product function category. For inorganic fertilizers, declared nutrients may include primary macronutrients, secondary macronutrients and micronutrients and several nutrients may be expressed in oxide rather than elemental forms. Oxide-based declarations such as P_2O_5, K_2O, CaO, MgO, Na_2O and SO_3 must therefore be converted before use in stoichiometric nutrient-solution calculations. Chloride is addressed separately from nutrient declaration; for example, the claim “poor in chloride” may only be used below 30 g kg^{-1} dry matter (European Parliament & Council of the European Union, 2019).

For this reason, the present work was restricted to single salts or mixtures of single salts with chemically defined composition. Such compounds can be represented directly by their molecular formulas, so that molar mass and stoichiometric composition are fully determined by the sum of the constituent elements and their stoichiometric coefficients. This ensures complete compositional transparency and avoids the uncertainty associated with incomplete commercial fertilizer declarations.

Many fertilizers used in greenhouse production are in fact single salts. A representative example is Epsom salt, $MgSO_4 \cdot 7H_2O$, marketed by Yara as YaraTera KRISTA MgS and declared as containing 16 % MgO and 32.5 % SO_3 (YARA, 2026). Chemically, pure $MgSO_4 \cdot 7H_2O$ contains 9.86 % Mg and 13.01 % S, corresponding to 16.35 % MgO and 32.48 % SO_3, or 38.98 % SO_4. Thus, 51.16 % of the compound mass consists of water of crystallization. Another example is calcium nitrate, which may occur as anhydrous $Ca(NO_3)_2$, as the hydrate $Ca(NO_3)_2 \cdot 4H_2O$, or as the double salt $5Ca(NO_3)_2 \cdot NH_4NO_3 \cdot 10H_2O$. These distinctions are stoichiometrically relevant because each form differs in molar mass and nutrient contribution, even where the commercial designation is similar.

2.3 Existing tools

A classical approach to nutrient solution calculation is the universal algorithm proposed by Sonneveld et al. (1999). The method follows an iterative balancing procedure in which fertilizer salts are added stepwise to meet nutrient targets, while the residual nutrient requirements are recalculated after each addition because each salt simultaneously contributes several ions. In this way, the final formulation is obtained through successive correction of remaining deficits under the constraints imposed by salt composition, target EC and raw-water composition.

Nutrient Solution Calculator is an Excel™-based tool developed by Luca Incrocci and supported by the EUPHOROS project (Pardossi et al., 2011). It was previously available for download via Wageningen University and was still accessible in 2024, but was no longer publicly available there in 2026 (Incrocci, 2007). A Spanish translation remains available online through the University of Almeria (Incrocci et al., 2020). The spreadsheet is designed to calculate the composition and cost of fertigation stock solutions (A+B) from three principal inputs: the ionic composition of the irrigation water, the crop nutrient recipe, including target pH, EC and nutrient concentrations and the technical characteristics of the fertigation system, particularly stock tank volume and dilution ratio. Furthermore, it evaluates the potential risk of salt precipitation in concentrated stock solutions. The spreadsheet is password-protected.

Another tool is HydroBuddy, a free and open-source software application developed by Daniel Fernández Pinto for the formulation of hydroponic nutrient solutions (Fernández Pinto, 2022). The program calculates the required quantities of fertilizers to achieve specified target nutrient concentrations. It includes a database of commonly used fertilizers whose compositions can be edited or extended by the user and allows the formulation of both direct nutrient solutions and concentrated stock solutions. The software determines the combination of fertilizer salts that best matches the desired nutrient composition through the solution of a system of linear equations and also provides an empirical estimation of the EC.

To the author's knowledge, no dedicated open-source R library is currently available for transparent, modular nutrient solution calculation in horticultural applications.

3 Material and Methods

The development of the computational framework followed a structured development and validation pipeline. The R package was developed using the devtools package (Wickham et al., 2025). The first step involved designing the underlying data structures. A compound-by-nutrient matrix was established in which rows represented fertilizer compounds (salts) and columns represented nutrients such as N, P, K and Ca. For solving, this matrix was transposed internally so that nutrient equations formed the rows used by the NNLS routine. Each fertilizer was parsed on the basis of its molecular formula into its elemental components using a database of molar masses and established stoichiometric rules.

Building on this foundation, an R package was developed. The package includes core functions for parsing chemical formulas and computing molar masses, as well as for converting between different chemical representations such as oxides and elemental forms (e.g., P_2O_5 to P or SO_4 to S). It also includes tools for transforming concentration units (e.g., from mmol L^{-1} to mg L^{-1}) using accurate molar-mass calculations. In addition, bidirectional transformations between compound-based and elemental quantities were integrated on the basis of stoichiometric relationships.

The central component of the package was the implementation of the nutrient solution formulation algorithm. This algorithm used `nnls()` to solve the system of equations under non-negativity constraints and to quantify deviations between the target and calculated solution in terms of absolute and relative error. The framework also allows users to assign weights to individual nutrients, thereby enabling the exploration of different formulation priorities and trade-offs.

The framework was evaluated through a series of reconstruction, benchmark and application use cases. These included reproducing crop-specific nutrient solutions, comparing the computed results with values reported in the literature and compiling crop- and author-specific nutrient solutions into a base library. The distribution of anions and cations across different solutions was visualised, for example, by ternary plots for cations (K^+, Ca^{2+}, Mg^{2+}) and anions (NO_3^-, SO_4^{2-}, PO_4^{3-}), in order to classify nutrient combinations systematically, as proposed by Steiner (1961).

3.1 Software Environment and Data Sources

3.1.1 R version, packages

The package `nutricalc` was written in R 4.5.1 (R Core Team, 2025) using Posit RStudio 2025.09.2+418 "Cucumberleaf Sunflower" (Posit team, 2025). The following packages were used: `stringr` (Wickham et al., 2025), `nnls` (Mullen & van Stokkum, 2024), `shiny` (Chang et al., 2025), `bslib` (Sievert et al., 2025), `Ternary` (Smith & Sanselme, 2025), `xml2` (Wickham et al., 2026). For the development in R `devtools` (Wickham, 2025) was used.

Code of the package is provided on GitHub under https://github.com/diolsa/nutricalc and the Shiny app is deployed under: https://nutricalc.shinyapps.io/nutricalc-app/

Code development was supported by AI-assisted LLM for debugging and optimisation to ensure code quality and computational reliability with OpenAI Codex (OpenAI, 2026b) and ChatGPT 5.4 (OpenAI, 2026a).

3.1.2 Built-in recipe datasets

Hewitt (1966) compiled a large proportion of the nutrient solution formulations used during the first half of the twentieth century. For the present work, selected standard recipes from this compilation were considered, particularly those that remain relevant in contemporary research and horticultural practice. In addition, 38 crop-specific horticultural recipes from Sonneveld & Straver (1994) were included. These recipes were assigned to the three groups `Olericulture`, `Floriculture` and `Fruticulture`. In the original source, Sonneveld & Straver (1994) also classified the formulations into categories, reflecting their practical and experimental suitability for the respective crop. The resulting set of formulations is shown in Table 3.1.

Table 3.1: Target compositions of crop-specific nutrient solutions from Sonneveld & Straver (1994), grouped by olericulture, floriculture and fruticulture.

		mmol L^{-1}							µmol L^{-1}						
Category	Recipe	NO_3–N	NH_4–N	P	K	Ca	Mg	S	Fe	Mn	Zn	B	Cu	Mo	Si
Olericulture	Endive in Recirculating Water	19	1.25	2	9	5	1.5	1.125	40	5	4	30	0.75	0.5	0
	Eggplant in Rockwool	15.5	1.5	1.25	6.75	3.25	2.5	1.5	15	10	5	30	0.75	0.5	0
	Eggplant in Rockwool (RDW)	11.75	1	1	6.5	2.25	1.5	1.125	15	10	5	20	0.75	0.5	0
	Bean in Rockwool	12	1	1.25	5.5	3.25	1.25	1.125	10	10	4	20	0.5	0.5	0
	Courgette in Rockwool	16	1.25	1.25	7.25	3.625	2	1.25	10	10	5	30	0.75	0.5	0
	Cucumber in Rockwool	16	1.25	1.25	8	4	1.375	1.375	15	10	5	25	0.75	0.5	750
	Cucumber in Rockwool (RDW)	11.75	1	1.25	6.5	2.75	1	1	15	10	5	25	0.75	0.5	750
	Propagation vegetable plants in rockwool	16.75	1.25	1.25	6.75	4.5	3	2.5	25	10	5	35	1	0.5	0
	Sweet Pepper in Rockwool	15.5	1.25	1.25	6.5	4.75	1.5	1.75	15	10	5	30	0.75	0.5	0
	Sweet Pepper in Rockwool (RDW)	12.75	1.25	1	5.75	3.25	1.125	1	15	10	4	25	0.75	0.5	0
	Lettuce in Recirculating Water	19	1.25	2	11	4.5	1	1.125	40	5	4	30	0.75	0.5	500
	Tomato in Rockwool	13.75	1.25	1.25	8.75	4.25	2	3.75	15	10	5	30	0.75	0.5	0
	Tomato in Rockwool (RDW)	10.75	1	1.25	6.5	2.75	1	1.5	15	10	4	20	0.75	0.5	0
Floriculture	Alstroemeria in Rockwool	11.25	1.25	1.25	6	2.875	1	1.25	25	10	4	25	0.75	0.5	0
	Alstroemeria in Rockwool (RDW)	7.5	0.75	1	4.75	2	0.75	1.25	25	5	4	20	0.75	0.5	0
	Anemone in Rockwool	13	1	1.5	6.5	3.75	1	1.25	35	5	4	30	0.75	0.5	0
	Carnation in Rockwool or Peat	13	1	1.25	6.25	3.75	1	1.25	25	10	4	30	0.75	0.5	0
	Carnation in Rockwool (reuse drainage)	7	0.75	0.8	4	1.625	0.6	0.7	20	5	3	20	0.5	0.5	0
	Anthurium andreanum in Rockwool or Peat	4.5	0.8	0.7	3	1	0.7	1	15	3	3	20	0.5	0.5	0
	Aster in Rockwool or Peat	13	1	1.25	6.25	3.75	1	1.25	25	10	4	30	0.75	0.5	0
	Bouvardia in Rockwool	13	1.25	1.75	6	4.25	1	1.5	25	5	3.5	20	0.75	0.5	0
	Bouvardia in Rockwool (RDW)	8	1	1.5	4	2.5	0.5	0.75	25	5	3.5	20	0.75	0.5	0
	Chrysanthemum in Recirculating Water	12.75	1.25	1	7.5	2.5	1	1	60	20	3	20	0.5	0.5	0
	Cymbidium in phenol foam	4.5	0.5	0.8	3	1.2	0.75	1.05	8	10	4	20	0.4	0.4	0
	Cymbidium in rockwool/urethane foam	4	1	0.8	2.8	1	0.75	1.25	8	10	4	20	0.4	0.4	0
	Euphorbia fulgens in Rockwool	11.5	1	1.5	6	3.5	1	1.5	35	10	3	20	0.5	0.5	0
	Freesia in Rockwool or Sand	14.5	1.25	1.25	7.75	3.375	1.5	1.5	25	10	4	25	0.75	0.5	0
	Gerbera in Rockwool	11.25	1.5	1.25	5.5	3	1	1.25	35	5	4	30	0.75	0.5	0
	Gerbera in Rockwool (RDW)	7.25	0.75	0.6	4.5	1.6	0.4	0.7	25	5	3	20	0.5	0.5	0
	Gypsophila in Rockwool	17	1.25	1.25	4	6	1.7	1.2	25	10	4	30	0.8	0.5	0
	Hippeastrum in Pumice	13	1	1.25	7.5	3.125	1	1.25	10	10	5	30	0.5	0.5	0
	Pot Plants in Expanded Clay	10.6	1.1	1.5	5.5	3	0.75	1	20	10	3	20	0.5	0.5	0
	Rose in Rockwool	11	1.5	1.25	4.5	3.25	1.125	1.25	25	5	3.5	20	0.75	0.5	0
	Rose in Rockwool (RDW)	4.3	0.85	0.5	2.15	0.9	0.5	0.5	15	5	3	15	0.5	0.5	0
	Statice in Rockwool	12	1	1	6	3	1	1	15	10	5	2.5	0.75	0.5	0
Fruticulture	Strawberry in Peaty Substrates	11.5	1	1	5.5	3.25	1.25	1.5	20	10	7	25	0.75	0.5	0
	Strawberry in Recirculating Water	10	0.5	1.25	5.25	2.75	1.125	1.125	20	10	4	20	0.75	0.5	0
	Melon in Rockwool	16.25	1	1.25	7.5	4.75	1.25	1.5	10	10	4	20	0.5	0.5	750

3.2 Data Structures and Chemical Representation

3.2.1 Elementary Entities

After the 2019 revision of the International System of Units, the base units are defined by fixed numerical values of fundamental constants (BIPM, 2019). The mole is the SI unit for the amount of a substance; its symbol is mol. The defining constant is the Avogadro constant, N_A. Thus:

$$1\,\text{mol} = 6.02214076 \times 10^{23}\ \text{elementary entities} \tag{3.1}$$

An elementary entity may be an atom, a molecule, or any other particle. In this work, the elementary entities are ions, as listed in Table 3.2. The molar masses were taken from the CRC Handbook of Chemistry and Physics (Haynes, 2017) and rounded to three decimal places. Not all listed elements are essential for plant nutrition. The set also includes non-essential and potentially toxic elements that may occur in the plant ionome or in fertilizers and source water, although not all of these elements were incorporated into the final framework.

Table 3.2: Molar masses of elementary entities used for nutrient formulation.

Symbol	Element name	Molar Mass [g mol^{-1}]
C	Carbon	12.011
O	Oxygen	15.999
H	Hydrogen	1.008
N	Nitrogen	14.007
P	Phosphorus	30.974
K	Potassium	39.098
Mg	Magnesium	24.305
S	Sulfur	32.065
Ca	Calcium	40.078
B	Boron	10.811
Cu	Copper	63.546
Cl	Chlorine	35.453
Fe	Iron	55.845
Mn	Manganese	54.938
Ni	Nickel	58.693
Zn	Zinc	65.380
Mo	Molybdenum	95.950
Na	Sodium	22.990
Al	Aluminum	26.982
Co	Cobalt	58.933
Se	Selenium	78.971
Si	Silicon	28.085

3.2.2 Fertilizer inventory

Table 3.3 lists the 40 fertilizer compounds implemented in the solver, including their chemical formulas, compound names and calculated molar masses. The molar mass of each compound was calculated from its chemical formula using the molar masses of the constituent elements and their stoichiometric coefficients reported in Table 3.2.

Table 3.3: Fertilizer compounds implemented in NutriCalc and their calculated molar masses.

Formula	Name	Molar mass [g mol^{-1}]
NH_4Cl	Ammonium chloride	53.492
$NH_4H_2PO_4$	Ammonium dihydrogen phosphate	115.025
$((NH_4)_6)Mo_7O_{24} \cdot 4\,H_2O$	Ammonium heptamolybdate tetrahydrate	1235.920
NH_4OH	Ammonium hydroxide	35.046
NH_4NO_3	Ammonium nitrate	80.043
$(NH_4)_2SO_4$	Ammonium sulfate	132.139
H_3BO_3	Boric acid	61.832
$CaCl_2 \cdot 2\,H_2O$	Calcium chloride dihydrate	147.014
$Ca(OH)_2$	Calcium hydroxide	74.092
$Ca(NO_3)_2$	Calcium nitrate	164.086
$Ca(NO_3)_2 \cdot 4\,H_2O$	Calcium nitrate tetrahydrate	236.146
$5\,Ca(NO_3)_2 \cdot NH_4NO_3 \cdot 10\,H_2O$	Calcium nitrate–ammonium nitrate decahydrate (5:1)	1080.623
$CaSO_4 \cdot 2\,H_2O$	Calcium sulfate dihydrate	172.169
$CuCl_2$	Copper(II) chloride	134.452
$CuSO_4 \cdot 5\,H_2O$	Copper(II) sulfate pentahydrate	249.682
$(NH_4)_2HPO_4$	Diammonium hydrogen phosphate	132.056
HCl	Hydrochloric acid	36.461
$C_{18}H_{16}N_2O_6FeNa$	Iron(III) sodium EDDHA	435.169
$C_{10}H_{12}N_2O_8FeNa \cdot 3\,H_2O$	Iron(III) sodium EDTA trihydrate	421.092
$Mg(NO_3)_2 \cdot 6\,H_2O$	Magnesium nitrate hexahydrate	256.403
$MgSO_4 \cdot 7\,H_2O$	Magnesium sulfate heptahydrate	246.471
$MnCl_2 \cdot 4\,H_2O$	Manganese(II) chloride tetrahydrate	197.904
$MnSO_4 \cdot H_2O$	Manganese(II) sulfate monohydrate	169.014
KH_2PO_4	Monopotassium dihydrogen phosphate	136.084
HNO_3	Nitric acid	63.012
H_3PO_4	Phosphoric acid	97.994
KCl	Potassium chloride	74.551
KOH	Potassium hydroxide	56.105
KNO_3	Potassium nitrate	101.102
K_2SiO_3	Potassium silicate	154.278
K_2SO_4	Potassium sulfate	174.257
$NaCl$	Sodium chloride	58.443
$NaOH$	Sodium hydroxide	39.997
$Na_2MoO_4 \cdot 2\,H_2O$	Sodium molybdate dihydrate	241.956
$NaNO_3$	Sodium nitrate	84.994
$Na_2B_4O_7 \cdot 10\,H_2O$	Sodium tetraborate decahydrate (borax)	381.367
H_2SO_4	Sulfuric acid	98.077
$ZnCl_2$	Zinc chloride	136.286
$ZnSO_4 \cdot 7\,H_2O$	Zinc sulfate heptahydrate	287.546
$ZnSO_4 \cdot 6\,H_2O$	Zinc sulfate hexahydrate	269.531

3.2.3 Compound-by-nutrient matrix

Table 3.4 shows the compound-by-nutrient matrix, listing the amount of each nutrient in mmol provided per mmol of compound for all 40 fertilizers. The table is organised with the chemical formula of each compound in the first column, followed by a column indicating the solver stage (1 for salts, 2 for acids and bases) and then columns for each nutrient. Chelates are also part of the full matrix, but are excluded from the table shown here for better readability. Each compound is additionally linked to metadata such as compound name, category and acid/base status.

Table 3.4: Compound–nutrient stoichiometric matrix used by the solver. Values indicate mmol nutrient supplied per mmol compound. Stage 1 = salts; Stage 2 = acids/bases.

Formula	Solver	NO3_N	NH4_N	P	K	Ca	Mg	S	Na	Cl	Fe	Mn	Zn	B	Cu	Mo	Si
$Ca(NO_3)_2$	1	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
NH_4NO_3	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
KH_2PO_4	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
KNO_3	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
K_2SO_4	1	0	0	0	2	0	0	1	0	0	0	0	0	0	0	0	0
$MgSO_4 \cdot 7\,H_2O$	1	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0
$CaCl_2 \cdot 2\,H_2O$	1	0	0	0	0	1	0	0	0	2	0	0	0	0	0	0	0
KCl	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0
$CaSO_4 \cdot 2\,H_2O$	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
$CuSO_4 \cdot 5\,H_2O$	1	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0
$(NH_4)_2SO_4$	1	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0
H_3BO_3	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
$MnSO_4 \cdot H_2O$	1	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0
$Na_2MoO_4 \cdot 2\,H_2O$	1	0	0	0	0	0	0	0	2	0	0	0	0	0	0	1	0
$ZnSO_4 \cdot 7\,H_2O$	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0
$NH_4H_2PO_4$	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
$Ca(NO_3)_2 \cdot 4\,H_2O$	1	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
$Mg(NO_3)_2 \cdot 6\,H_2O$	1	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
$C_{10}H_{12}N_2O_8FeNa \cdot 3\,H_2O$	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0
$ZnSO_4 \cdot 6\,H_2O$	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0
$Na_2B_4O_7 \cdot 10\,H_2O$	1	0	0	0	0	0	0	0	2	0	0	0	0	4	0	0	0
$C_{18}H_{16}N_2O_6FeNa$	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0
K_2SiO_3	1	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	1
$NaCl$	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0
$(NH_4)_2HPO_4$	1	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0
$CuCl_2$	1	0	0	0	0	0	0	0	0	2	0	0	0	0	1	0	0
$((NH_4)_6)Mo_7O_{24} \cdot 4\,H_2O$	1	0	6	0	0	0	0	0	0	0	0	0	0	0	0	7	0
$ZnCl_2$	1	0	0	0	0	0	0	0	0	2	0	0	1	0	0	0	0
$MnCl_2 \cdot 4\,H_2O$	1	0	0	0	0	0	0	0	0	2	0	1	0	0	0	0	0
NH_4Cl	1	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0
$NaNO_3$	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
$5\,Ca(NO_3)_2 \cdot NH_4NO_3 \cdot 10\,H_2O$	1	11	1	0	0	5	0	0	0	0	0	0	0	0	0	0	0
HNO_3	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
KOH	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
H_2SO_4	2	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
HCl	2	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
H_3PO_4	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
$NaOH$	2	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
NH_4OH	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$Ca(OH)_2$	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

3.3 Algorithmic Methods

3.3.1 Formula parsing and molar mass calculation

Molecular formulas were treated as the primary formal representation of fertilizer compounds within the framework. Parsing therefore began by tokenizing each formula into element symbols, integer multipliers, bracketed groups and optional hydrate components, for example the `·4H2O` term in hydrated salts. After structural validation, grouped expressions were expanded to absolute atom counts for each element. Molar mass was then calculated as the weighted sum of elemental contributions, such that for each element i, the stoichiometric coefficient n_i was multiplied by its tabulated atomic molar mass M_i, giving:

$$M = \sum_i n_i M_i$$

This step is fundamental to the framework, because all downstream stoichiometric conversions between compound amounts and nutrient-element amounts depend directly on the resolved elemental composition and the corresponding molar masses. For example, the molar mass of calcium nitrate tetrahydrate was computed as follows:

```
compute_molar_mass("Ca(NO3)2·4H2O")
```

```
[1] 236.146
```

The returned value corresponds to a molar mass of 236.146 g mol^{-1} for $Ca(NO_3)_2 \cdot 4H_2O$.

3.3.2 Stoichiometric conversions

Numerically, the official SI unit for the substance concentration is mol m^{-3}; however, this unit is equivalent to mmol L^{-1}. Therefore, the following relationships for conversion between mmol L^{-1} and mg L^{-1} were applied, following Appelo and Postma (Appelo, 2005):

$$\begin{aligned} \text{mg L}^{-1} &= \text{mmol L}^{-1} \times \text{M} \\ \text{mmol L}^{-1} &= \frac{\text{mg L}^{-1}}{\text{M}} \end{aligned}$$

where

M is the molar mass of the corresponding element $[\text{g mol}^{-1}]$.

Within the library, conversion between mmol L^{-1} and mg L^{-1} as elements can be accomplished with the `convert_units()` function. When the input is provided as a named numeric vector, the names are interpreted as nutrient identifiers (e.g., `Ca`, `P`, `NO3_N`) and are internally mapped to their corresponding element symbols to select the appropriate molar mass (from element_molar_mass_df) for conversion. For example, converting a complete set of recipe values from mg L^{-1} to mmol L^{-1}:

```
convert_units(c(Ca = 211, P = 46), to = "mmol/L")
```

```
      Ca        P
5.264734 1.485117
```

The conversion can also be applied to a single scalar concentration. In this case, the element must be specified explicitly, since a scalar input carries no nutrient name from which the element can be inferred. The element may be supplied either as a named argument or positionally in the order `(x, element, to)`:

```
convert_units(x = 15, element = "N", to = "mg/L")
```

```
[1] 210.105
```

When concentrations are declared in mg L^{-1}, masses are often reported in oxidized or compound forms (e.g. SO_4 or MgO). To convert between such compound-based representations and their elemental equivalents, the `converte()` function can be used:

$$x_{\mathrm{to}} = x_{\mathrm{from}} \times \frac{M_{\mathrm{to}}}{M_{\mathrm{from}}} \times \frac{\nu_{from,E}}{\nu_{to,E}}$$

where

x_{from} is the quantity expressed as the source formula (input value),

x_{to} is the equivalent quantity expressed as the target formula (output value),

M_{from} is the molar mass of the source formula $[\mathrm{g\,mol^{-1}}]$,

M_{to} is the molar mass of the target formula $[\mathrm{g\,mol^{-1}}]$,

$\nu_{from,E}$ is the number of atoms of the anchor element E in the source formula [dimensionless],

$\nu_{to,E}$ is the number of atoms of the anchor element E in the target formula [dimensionless].

```
converte("SO4", "S")
```

```
[1] 0.3337983
```

It is also possible to specify an explicit quantity to be converted, for example when converting a given mass concentration

```
converte("N", "NO3", 5)
```

```
[1] 22.13322
```

If multiple elements are shared between the source and target formulas, the anchor element E must be specified to avoid ambiguity.

```
converte("(NH4)2HPO4", "NH4H2PO4", element = "P")  # P as the anchor
```

```
[1] 0.871032
```

3.3.3 NNLS optimisation with weighting

NNLS is a numerical optimisation algorithm for solving linear least-squares problems under non-negativity constraints. It was written in 1973 in FORTRAN 77 by Lawson and Hanson at the Jet Propulsion Laboratory and published in their book *Solving Least Squares Problems* (Lawson & Hanson, 1974, 1995). In R, the Lawson–Hanson algorithm is implemented by the package `nnls` (Mullen & van Stokkum, 2024).

For nutrient solution formulation, the objective is to determine the amounts of salts that fit a specified target nutrient composition as closely as possible. Each salt contributes fixed stoichiometric amounts of ions, so the relationship between the amount of each salt and the resulting nutrient concentration in solution is linear. Because salts can only be added and not removed, their amounts are constrained to be non-negative. Nutrient solution design can therefore be formulated as a linear mixture fitting problem subject to non-negativity constraints, which can be solved using the NNLS algorithm.

Because many fertilizer salts supply multiple nutrients simultaneously exact matching of all nutrient targets is rarely possible. The optimisation therefore identifies the best feasible compromise among competing nutrient constraints.

Assume that n fertilizer salts are available and m nutrients are considered.

In `NutriCalc`, fertilizer stoichiometry is stored in a matrix

$$S \in \mathbb{R}^{n \times m},$$

where rows correspond to salts ($j = 1, \dots, n$) and columns correspond to nutrients ($i = 1, \dots, m$). The entry S_{ji} gives the amount of nutrient i supplied by one unit of salt j (mmol nutrient per mmol salt).

Let

- $x \in \mathbb{R}^n_{\geq 0}$ denote the vector of salt amounts (mmol L^{-1}),

- $b \in \mathbb{R}^m$ denote the vector of target nutrient concentrations (mmol L^{-1}).

For fitting, the solver uses the transposed stoichiometry matrix

$$A = S^\top \in \mathbb{R}^{m \times n},$$

so that rows correspond to nutrients and columns correspond to salts. The system therefore consists of m nutrient equations and n non-negative decision variables. Predicted (achieved) nutrient concentrations are

$$\hat{b} = Ax.$$

Thus, each row of A represents the linear contribution of all salts to one nutrient and each column represents the nutrient composition of one salt. The optimisation therefore selects salt amounts so that the combined nutrient profile best approximates the desired composition.

3.3.4 Target scaling and importance weighting

`NutriCalc` does not solve the unweighted least-squares problem $\min_{x \geq 0} \|Ax - b\|_2$. Instead, residuals are rescaled relative to target magnitude and weighted according to user-defined nutrient importance.

Conceptually, the solver minimises the total weighted squared deviation between achieved and target nutrient concentrations. For nutrients with nonzero targets, deviations are measured relative to the target (i.e. proportional error). For nutrients with zero targets, deviations are measured in absolute concentration units.

This relative scaling is necessary because nutrient concentrations often differ by several orders of magnitude (for example, macronutrients versus micronutrients). Without scaling, nutrients present at large absolute concentrations would dominate the optimisation and prevent meaningful fitting of low-concentration nutrients.

Scaling and importance weighting operate on residuals rather than on concentrations themselves. optimisation decisions therefore depend on deviations from targets rather than absolute concentration levels.

3.3.4.1 Importance weighting of nutrient targets

In practical nutrient formulation, not all targets must be matched with the same precision. Some nutrients are critical for plant performance and must be maintained close to their target concentration, whereas others can vary within a broader range without adverse effects. The optimisation therefore allows nutrient-specific importance levels that determine how strongly deviations from individual targets are penalized.

User-defined importance values I_i express the desired strictness of matching each nutrient. Higher importance means that deviations from the target are penalized more strongly, forcing the solver to prioritize matching that nutrient even if this leads to larger deviations in less important nutrients.

Importance values are restricted to the interval $[-2, 2]$ and converted into positive numerical weights using an exponential mapping:

$$\tilde{I}_i = \min(2, \max(-2, I_i)),$$

$$w_i = 0.1 \cdot 1000^{\tilde{I}_i}.$$

Each unit increase in importance multiplies the penalty on deviations by a factor of 1000. This creates clear separation between importance levels and ensures that highly important nutrients strongly influence the optimisation outcome. This mapping produces weights spanning approximately 10^{-7} to 10^5.

The weights are incorporated into the least-squares objective using the diagonal matrix

$$W = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_m}),$$

which scales the residuals before minimization. The optimisation therefore minimises a weighted sum of squared deviations, where each nutrient contributes proportionally to its assigned importance.

3.3.4.2 Target-based scaling of residuals

Nutrient targets differ substantially in magnitude, with macronutrients typically present at concentrations several orders of magnitude larger than micronutrients. If deviations were measured only in absolute units (mmol L^{-1}), nutrients with large target concentrations would dominate the optimisation and small but biologically important deviations in low-concentration nutrients would have little influence on the solution.

To avoid this imbalance, residuals are scaled relative to target magnitude. For each nutrient i, define a scaling factor

$$s_i = \begin{cases} b_i, & b_i \neq 0, \\ 1, & b_i = 0. \end{cases}$$

For nutrients with nonzero targets, deviations are therefore measured relative to the target value. For nutrients with zero targets, deviations are measured in absolute concentration units.

To apply this scaling, each nutrient equation is divided by its corresponding factor s_i. In matrix form, the target-scaled design matrix is

$$A_{\text{sub}} = \text{diag}(1/s)\, A,$$

and the scaled target vector is b/s, where division is performed elementwise.

The NNLS problem is then

$$\min_{x\geq 0} \|W\,(A_{\text{sub}}x - b/s)\|_2^2\,.$$

This is equivalent to minimizing a weighted sum of squared scaled residuals. For nutrients with nonzero targets, the optimisation minimises weighted relative deviations; for nutrients with zero targets, it minimises weighted absolute deviations. The scaling ensures that all nutrients contribute comparably to the optimisation regardless of their concentration range, while importance weights control how strictly individual targets are enforced.

Operationally, the solver predicts nutrient concentrations from candidate salt amounts, computes deviations from the targets, rescales and weights these deviations according to target magnitude and nutrient importance and determines the non-negative salt combination that minimises the resulting weighted least-squares objective.

3.3.5 Interpretation of the solution

The NNLS solver returns the optimal non-negative salt amounts x. Because the optimisation is performed on scaled and weighted equations, achieved nutrient concentrations are recomputed using the original, unscaled stoichiometry:

$$\hat{b} = Ax.$$

The vector x therefore represents the physically meaningful fertilizer composition and is interpreted directly as mmol L^{-1} of each salt to add to the nutrient solution. These values can therefore be used directly to prepare the nutrient solution.

3.3.6 Error quantities reported by `NutriCalc`

After computing achieved nutrient concentrations, deviations from the target composition are quantified using several complementary error measures. These metrics describe both absolute and relative discrepancies between predicted and desired nutrient levels.

The absolute error for nutrient i is

$$e_i = \hat{b}_i - b_i,$$

expressed in mmol L^{-1}. This measures the direct concentration difference between achieved and target values.

Percent error expresses the deviation relative to the target concentration:

$$p_i = 100 \cdot \frac{e_i}{b_i}.$$

Percent error is computed elementwise. Non-finite values (e.g. division by zero) are recorded as missing. If $b_i = 0$ and $|\hat{b}_i| < 10^{-9}$, the percent error is set to 0.

To summarise overall deviation across all nutrients, sum of squared error is computed as

$$\mathrm{SSE} = \sum_{i=1}^{m} e_i^2.$$

This metric reflects the total magnitude of absolute concentration differences.

Relative squared error evaluates deviations relative to target magnitude. Relative residuals are defined as

$$r_i = \frac{e_i}{b_i},$$

and non-finite values are omitted. The aggregate measure is reported as the sum of squared relative errors

$$\mathrm{SSRE} = \sum_{i=1}^{m} r_i^2.$$

This metric summarises proportional deviations across nutrients and provides a scale-independent measure of solution accuracy. Nutrients with zero target concentration were excluded from relative error metrics.

3.3.7 Two-stage NNLS solver: separation of neutral salts and acid–base reagents

In addition to the standard optimisation, `NutriCalc` provides an optional two-stage solver that incorporates acid–base reagents. These compounds are included in the global stoichiometry matrix and distinguished from neutral fertilizer salts using the classification `salt_info$is_acid_base`. The purpose of this separation is to ensure that nutrient targets are satisfied using neutral salts first and whenever possible, while acids and bases are used only to correct remaining deficits that neutral salts cannot resolve. This prevents acid–base reagents from acting as unconstrained degrees of freedom in the primary optimisation and yields chemically more interpretable fertilizer compositions.

The two-stage procedure is activated only when the acid/base option is enabled. When this option is disabled, acid/base compounds are excluded from the selected compound set and nutrient optimisation is performed using neutral salts only. Boric acid and potassium silicate are classified in the neutral group because, in this implementation, they are treated as nutrient sources rather than as pH-active reagents.

In the first stage, NNLS optimisation is performed using only compounds classified as neutral salts. This stage uses the same weighted and target-scaled objective as the standard solver and produces achieved nutrient concentrations

$$\hat{b}^{(1)}.$$

Residual nutrient deviations are then computed as

$$r = b - \hat{b}^{(1)}.$$

Residual entries are filtered before any second-stage correction. Deviations within the relative tolerance are treated as negligible and set to zero. Residuals are also set to zero for nutrients that cannot be supplied by any acid or base (non-fixable nutrients). Negative residuals (oversupply) are set to zero so that only undersupplied nutrients remain eligible for correction.

If the two-stage mode is enabled and at least one fixable nutrient exceeds the tolerance criterion defined by

$$r_i > \max(\texttt{tol\_abs}, \texttt{tol\_rel}\, b_i),$$

a second NNLS optimisation is performed using only acid/base compounds. The target for this stage is the filtered residual vector from stage 1. Scaling and importance weighting are applied using the same rules as in the standard solver, but computed from the residual target rather than the original nutrient targets. Stage 2 therefore supplies only remaining nutrient deficits that neutral salts could not provide.

Compound amounts from stage 1 and stage 2 are then inserted into their original compound indices to form the complete fertilizer vector (x). Final nutrient concentrations are recomputed from the full unscaled stoichiometry matrix,

$$\hat{b} = Ax,$$

and all error metrics are evaluated from these final concentrations.

Accordingly, the two-stage solver acts as a conditional refinement: neutral salts determine the primary nutrient composition and acid–base reagents are introduced only when residual deficits exceeding the tolerance criterion remain after the neutral fit. Lowering the importance of a specific nutrient reduces the penalty for deviating from its target during stage 1, allowing a residual deficit to remain primarily in that nutrient while the remaining nutrients are matched more closely. Because stage 2 optimisation operates only on residual undersupply, this deficit can then be supplied during the second stage by the corresponding acid–base compounds.

3.4 Chemical Modeling

3.4.1 Water quality and input handling

Nutrient formulation was implemented using a target-based optimisation workflow in NutriCalc. Users specify desired final nutrient concentrations as a nutrient target vector. The solver then computes the fertilizer salt additions needed to match this target. This approach is exact only when the initial water matrix is essentially deionised or demineralised water. In practical cultivation systems, however, source water already contains dissolved ions that contribute to the final solution and must therefore be accounted for before fertilizer optimisation.

To represent this baseline contribution, water chemistry is modelled as a water vector containing pre-existing ion concentrations. The formulation step uses an adjusted target vector obtained by subtracting the baseline water vector from the desired nutrient target vector. For each nutrient ion i, the solver input is defined as:

$$T_i^{adj} = \max\Big(T_i^{target} - W_i,\ 0\Big)$$

where T_i^{target} is the desired final concentration and W_i is the concentration already present in the source water. Negative adjusted concentrations are not physically meaningful because fertilizer salts can only be added. Therefore, adjusted targets are bounded below by zero. The lower bound at zero reflects the additive-only fertilizer model used in the optimizer; excess baseline ions cannot be removed within this step. The resulting adjusted vector is then passed to the salt optimizer.

In addition to ionic concentrations, two buffering descriptors are important for water characterization: acid capacity (Säurekapazität, KS 4.3) and base capacity (Basenkapazität, KB 8.2). These are not treated as primary fertilizer targets, but are retained as water properties because they influence downstream interpretation of pH behavior.

Because users may operate with blended feed waters, NutriCalc supports mixing of multiple source waters prior to target subtraction. Each source is represented as a separate composition vector and the composite baseline is computed by mass balance. For two source waters with compositions W_1 and W_2 and volumes V_1 and V_2, the mixed baseline composition is:

$$W^{mix} = \frac{V_1 W_1 + V_2 W_2}{V_1 + V_2}.$$

This expression is evaluated component-wise for all dissolved constituents of the water vector. The adjusted nutrient target is subsequently computed as:

$$T^{adj} = \max\big(T^{target} - W^{mix},\ 0\big)\,.$$

This enables realistic handling of scenarios such as blending municipal water with deionised water, rainwater, or other process waters before nutrient dosing.

Water quality inputs were derived from publicly available authority datasets where possible. For Berlin, NutriCalc retrieves supplier data and maps it to the internal water-vector representation. In practice, users can query location-specific baseline water data by postal code and optionally modify values to reflect local measurements or temporal variability.

Overall, this method ensures that fertilizer optimisation is performed against the true chemical starting point rather than an idealized zero-background assumption. By explicitly modeling baseline composition, optional source-water mixing and buffering-related capacities, the resulting formulations are more representative of real cultivation solutions and suitable for subsequent pH and EC evaluation.

3.4.2 Estimation of pH via charge balance

Solution pH is calculated using `ph_from_achieved()` based on the achieved nutrient composition returned by the solver (mmol L^{-1}). The calculation is performed as a charge-balance problem in which pH is treated as an unknown variable. The hydrogen ion concentration is determined such that the aqueous solution satisfies electroneutrality, expressed in milliequivalents per liter (meq L^{-1}), such that total positive and negative charge are equal (Appelo, 2005).

The calculation is formulated as a root-finding problem in which the charge-balance residual is expressed as a function of pH. The root variable is defined as

$$\mathrm{pH}_c = -\log_{10}([\mathrm{H}^+]).$$

For each trial value of pH_c, equilibrium speciation is computed and the resulting charge imbalance is evaluated. The solution pH corresponds to the value of pH_c for which the charge residual equals zero within numerical tolerance.

Strong ions (e.g., K^+, Na^+, Ca^{2+}, Mg^{2+}, NO_3^-, Cl^-) contribute with fixed concentrations from the achieved nutrient composition, whereas variable ions (e.g., H^+, OH^-, NH_4^+) are determined from equilibrium speciation. Weak acid–base systems (e.g., sulfate, phosphate, carbonate, borate, silicate) are included in the speciation model and contribute to the charge balance according to their equilibrium distributions.

Carbonate speciation is parameterized either from a carbonate-system input treated as total inorganic carbon when provided (e.g., `KS4_3`, interpreted approximately as HCO_3^-), or from fixed dissolved $CO_2(aq)$ when provided (e.g., `KB8_2`, interpreted approximately as CO_2). Optional chelating ligands (e.g., for Fe) are included as fixed-charge anions when ligand totals are present (default net charges: EDTA, EDDHA, HBED $z = -4$ and DTPA $z = -5$).

Acid–base equilibria are evaluated using thermodynamic dissociation constants at 25 °C (Haynes, 2017). The equilibrium systems included in the speciation model are summarized in Table 3.5.

Table 3.5: Acid–base equilibrium systems included in the pH speciation model.

System	Equilibrium species
Water	H^+, OH^-
Ammonia	NH_4^+, NH_3
Sulfate	HSO_4^-, SO_4^{2-}
Phosphate	H_3PO_4, $H_2PO_4^-$, HPO_4^{2-}, PO_4^{3-}
Carbonate	$CO_2(aq)$, HCO_3^-, CO_3^{2-}
Borate	$B(OH)_3$, $B(OH)_4^-$
Silicate	H_4SiO_4, $H_3SiO_4^-$

Non-ideal solution behavior is represented using Davies activity coefficients at 25 °C, using the Debye–Hückel parameter for water ($A = 0.5085$). Ionic strength is computed as

$$I = \frac{1}{2}\sum c_i z_i^2.$$

For each trial pH_c, ionic strength and activity coefficients are determined iteratively because they depend on the speciation they influence. Ionic strength is computed from species concentrations, activity coefficients are updated from ionic strength and speciation is recalculated. This fixed-point iteration is repeated until successive estimates converge. The resulting activity coefficients are then applied to all equilibrium expressions.

After convergence of the inner speciation loop, the charge residual is evaluated as total positive equivalents minus total negative equivalents. Positive charge includes fixed cations and variable H^+ and NH_4^+. Negative charge includes fixed anions, OH^- and the equivalent contributions of sulfate, phosphate, carbonate, borate and silicate species, together with optional fixed chelate charge.

The bisection search is initialized over a bounded pH interval (default 3–9), reflecting the typical range of horticultural nutrient solutions and improving numerical stability. If the charge-balance function is not bracketed within this interval, the search range is expanded automatically.

The reported pH is activity-based,

$$\mathrm{pH} = -\log_{10}\big(\gamma_H [\mathrm{H}^+]\big),$$

and therefore may differ slightly from pH_c. All calculations assume 25 °C and exclude precipitation reactions and explicit metal–ligand complexation beyond optional fixed-charge chelate terms.

3.4.3 EC estimation using ionic diffusion coefficients

Electrical conductivity (EC) quantifies the ability of an aqueous solution to transport electrical charge and is therefore determined by the concentration and mobility of dissolved ions. Several approaches exist for estimating EC from chemical composition, ranging from simple empirical relationships to ion-specific physicochemical models.

In practical greenhouse fertilization, EC is often estimated from the total ionic charge concentration expressed in milliequivalents per liter (meq L^{-1}). A widely used rule of thumb for greenhouse nutrient solutions is (van der Lugt et al., 2020):

$$\text{EC (mS cm}^{-1}) = \frac{\sum \text{meq cations} + \sum \text{meq anions}}{20}.$$

Because aqueous solutions satisfy electroneutrality,

$$\sum \text{meq cations} \approx \sum \text{meq anions},$$

this relationship can also be interpreted as a proportionality between EC and total ionic charge concentration. A closely related approximation for dilute aqueous solutions is described by Appelo (2005), where conductivity in µS cm^{-1} is approximately proportional to total milliequivalents per liter (typically valid below about 1.5 mS cm^{-1}):

$$\sum \text{meq ions} \approx \frac{\text{EC } (\mu\text{S cm}^{-1})}{100}.$$

More refined empirical relationships have been proposed for greenhouse nutrient solutions with relatively constrained ionic composition. Based on experimental calibration, Sonneveld & Straver (1994) proposed the following regression relating EC to total cation concentration:

$$\text{EC} = 0.095 \cdot \sum \text{meq cations} + 0.19,$$

where EC is expressed in mS cm^{-1}.

Although such empirical formulations may improve predictive accuracy within their calibration range, they do not explicitly account for differences in ionic mobility among individual species. Because the ionic species present in solution are known in `NutriCalc`, EC can be estimated mechanistically from their concentration, charge and mobility. The limiting molar ionic conductivity can be related to the diffusion coefficient through the Nernst–Einstein relation (Kalka, 2020). In `NutriCalc`, this relation is extended by a mobility correction term accounting for non-ideal solution behavior. This method is used in hydrochemical programs such as PHREEQC (up to version 3.8) and AQION 8.7.15 (Appelo, 2010; Kalka, 2020).

$$\mathrm{EC} = \left(\frac{F^2}{RT}\right) \sum_i D_i z_i^2 \left(\gamma_i\right)^{\alpha_i} c_i,$$

where

$$\begin{aligned}
F &\text{ is Faraday's constant } [9.6485 \times 10^4 \text{ C mol}^{-1}],\\
R &\text{ is the gas constant } [8.31446 \text{ J mol}^{-1}\text{ K}^{-1}],\\
T &\text{ is absolute temperature } [T = t_\mathrm{C} + 273.15 \text{ K}],\\
D_i &\text{ is the diffusion coefficient of ion } i \text{ } [\mathrm{m}^2\text{ s}^{-1}],\\
z_i &\text{ is the ionic charge number},\\
\gamma_i &\text{ is the activity coefficient of ion } i,\\
\alpha_i &\text{ is an empirical mobility scaling exponent},\\
c_i &\text{ is the molar concentration of ion } i.
\end{aligned}$$

Concentrations used in the conductivity summation are expressed in mol m^{-3}. Concentrations obtained in mmol L^{-1} are numerically identical to mol m^{-3} and are therefore used directly without conversion. For conductivity, concentrations are expressed in mol m^{-3} whereas ionic strength for activity modeling is calculated from concentrations expressed in mol L^{-1}. Diffusion coefficients in the internal data tables are stored in units of 10^{-5} cm^2 s^{-1} at 25 °C and converted to SI units prior to calculation:

$$1 \times 10^{-5} \text{ cm}^2\text{ s}^{-1} = 1 \times 10^{-9} \text{ m}^2\text{ s}^{-1}.$$

The activity of a dissolved species depends not only on its concentration but also on the ionic strength of the solution (Amelung et al., 2018). Ionic strength (I) is calculated for activity modeling from molar concentrations expressed in mol L^{-1}:

$$I = \frac{1}{2} \sum_i z_i^2 c_i,$$

Because the classical Nernst–Einstein relation applies strictly at infinite dilution, a mobility scaling factor is introduced to account empirically for inter-ionic interactions at finite ionic strength as described by Appelo (2010). The scaling exponent is defined as

$$\alpha_i = \begin{cases} \frac{0.6}{\sqrt{|z_i|}}, & I \leq 0.36|z_i| \\ \frac{\sqrt{I}}{|z_i|}, & I > 0.36|z_i| \end{cases}$$

providing an ionic-strength-dependent attenuation of individual ionic contributions.

Activity coefficients (γ_i) are obtained using the selected activity model implemented in `NutriCalc`. Two models are currently implemented: the limiting Debye–Hückel model and the Davies model (Kalka, 2023). In the limiting Debye–Hückel model, the activity coefficient is given by

$$-\log_{10}(\gamma_i) = A z_i^2 \sqrt{I},$$

whereas in the Davies model it is given by

$$-\log_{10}(\gamma_i) = A z_i^2 \left(\frac{\sqrt{I}}{1+\sqrt{I}} - 0.3I \right),$$

where $A = 0.5085$ at 25 °C, z_i is the ionic charge number and I is the ionic strength in mol L^{-1}.

The limiting Debye–Hückel formula is typically applied at ionic strengths up to 0.1 mol L^{-1} whereas the Davies formula is generally considered applicable up to about 0.5 mol L^{-1}. Currently the Davies activity correction is used as the default in the `NutriCalc` Shiny app.

The calculation is applied to the fully speciated aqueous ion composition obtained after nutrient formulation and equilibrium pH calculation. All dissolved charged species, including hydrogen and hydroxide ions, contribute to the conductivity sum according to their diffusion properties and concentrations.

Diffusion coefficients of ionic species at 25 °C were retrieved from Haynes (2017). The approach assumes fully dissociated strong electrolytes and does not account for neutral species or metal–ligand complexes. Consequently, the model provides a physically based estimate of conductivity that resolves species-specific contributions but may deviate from measured EC where complexation, ion pairing or strong ion association is significant.

The osmotic potential is estimated from the calculated EC value using the empirical relationship $\Psi_{\mathrm{osm}} = -36 \cdot \mathrm{EC}$, where Ψ_{osm} is expressed in kPa and EC in mS cm^{-1}. This relationship is given by George et al. (2012) based on United States Salinity Laboratory Staff (1954), where it was originally formulated as a relation between osmotic potential and the electrical conductivity of the saturation extract.

3.5 Shiny Application Workflow

The Shiny application was developed as an interactive interface to the computational workflow described in the previous sections. The purpose of the following section is therefore not only to document the user interface, but also to make the case-study workflows reproducible by linking the visible input and output elements of the application to the corresponding computational steps.

3.5.1 Installation or launching the web application

To install and run the application locally, the package `nutricalc` can be installed from GitHub using the package `remotes` and the following code:

```
install.packages("remotes")
remotes::install_github("diolsa/nutricalc")
library(nutricalc)
run_app()
```

Alternatively, the deployed Shiny application can be accessed online without local installation at https://nutricalc.shinyapps.io/nutricalc-app/.

Once launched, the application opens with the title `NutriCalc: Nutrient Solution Formulation in Horticultural Sciences`.

3.5.2 Input parameters

The left-hand panel of the Shiny graphical user interface (GUI), titled `Nutrient Targets & Fertilizers`, comprises five tabs: `Nutrient Targets`, `Water`, `Fertilizers`, `Recipes` and `Tissue`. These tabs expose the inputs that are mapped to target vectors, source-water vectors and solver constraints in the optimisation workflow.

3.5.2.1 Nutrient Targets

Within the `Nutrient Targets` tab, each nutrient is represented by three user-editable controls: a `Target` input field, a `Water` input field and a `Priority` selector. The `Target` field defines the desired concentration in the final nutrient solution, whereas the `Water` field specifies the concentration contributed by the source water. Input values can be displayed and entered in $\mathrm{mmol\,L^{-1}}$, $\mathrm{\mu mol\,L^{-1}}$, or $\mathrm{mg\,L^{-1}}$. In addition, alkalinity-related water parameters can be entered using the `KS 4.3` and `KB 8.2` fields. The `Priority` control is implemented as a slider ranging from −2 to +2 and specifies the relative importance of each nutrient during optimisation. After unit conversion and water correction, these values define the target vector and weighting used by the optimisation routine.

At the bottom of the panel, several global controls are available for workflow management. The `Set all to 0` button resets all target and water input values to zero, whereas `Reset to defaults` restores the predefined default settings. The `Use acids/bases` option activates the secondary optimisation stage after completion of the initial solver step, thereby allowing the inclusion of acids and bases for further adjustment. Finally, the `Result` button initiates the solver and opens the calculated formulation output.

3.5.2.2 Water

The `Water` tab is used to define and modify the source-water composition. Mean water-analysis values for Berlin postal code areas (`PLZ`) from BWB (2026) can be retrieved by entering a valid postal code (in the range from `10115` to `14199`). Retrieved values can be transferred to the `Water` input fields in the `Nutrient Targets` tab or into the water-mixing interface as `Water 1`.

Within the mixing interface, two separate source waters, `Water 1` and `Water 2`, can be defined. Concentrations for both sources can be entered manually. If one of the two sources is left empty, it is assumed to be deionised water. A slider is used to specify the mixing ratio between `Water 1` and `Water 2`. In this way, blends of different water sources, such as tap water, drainage water, rainwater, aquaculture effluent, or other custom nutrient-containing solutions, can be simulated. The resulting mixture can be transferred to the `Water` input fields in the `Nutrient Targets` tab. These inputs define the source-water vector that is incorporated into the final solution and subtracted from the nutrient targets before optimisation.

3.5.2.3 Fertilizers

The `Fertilizers` tab is used to define the fertilizer salts, acids and bases available for optimisation. Compounds are displayed in a list together with their chemical formulas and names. At the top of the tab, a `Salt search` text field is provided to filter the list. Both chemical formulas and compound names can be entered and the visibility of the listed compounds updates dynamically according to the search term. A `Clear search` button resets the search input and restores the full list.

Compounds can be selected or deselected manually using the checkbox list. In addition, the `Select all visible` and `Deselect all visible` buttons allow batch selection or deselection of compounds currently shown in the filtered list. Thus, these controls operate on the subset of compounds visible under the active search term rather than on the complete compound list. The selected compounds determine the subset of the compound-by-nutrient matrix that is passed to the NNLS solver. Below the list, a status line provides summary information in the form `Using x selected of y visible (total z)`, where `x` denotes the number of currently selected compounds, `y` the number of currently visible compounds and `z` the total number of available compounds.

3.5.2.4 Recipes

The Recipes tab provides a collection of crop-specific and standard nutrient solutions used in horticulture. It is divided into four categories: Olericulture, Fruticulture, Floriculture and Standard Formulations. Most recipes in the first three categories are based on Sonneveld & Straver (1994), whereas the fourth category contains formulations compiled from Hewitt (1966). The compilation by Hewitt (1966) represents one of the most comprehensive collections of hydroponic nutrient solution formulations published in the twentieth century.

Once a Category is selected, the corresponding Recipe list is displayed. Selecting a recipe transfers predefined nutrient targets to the Target input fields in the Nutrient Targets tab. For recipes reported together with specific fertilizer salts, selection of the recipe can also modify the set of compounds selected in the Fertilizers tab. In addition, recipes are stored in different concentration units, with some reported in $\mathrm{mmol\,L^{-1}}$ and others in $\mathrm{mg\,L^{-1}}$.

3.5.2.5 Tissue

The Tissue tab allows formulation of a nutrient solution based on the mass-balance approach described by Langenfeld et al. (2022). Tissue nutrient concentrations can be entered on a dry-mass basis and displayed in either % or ppm. In the central part of the tab, water-use efficiency (WUE) is calculated as dry mass divided by transpired water, with dry mass entered in g and transpired water in L. The resulting WUE value is displayed below the corresponding input fields.

Based on tissue nutrient concentration and WUE, nutrient demand per unit of transpired water is estimated and converted into nutrient targets using the mass-balance approach. In addition to the original approach of Langenfeld et al. (2022), total nitrogen is partitioned into nitrate and ammonium in accordance with the principle of electroneutrality. This extension is discussed in a later section.

3.5.3 Output parameters

The right-hand side of the Shiny graphical user interface (GUI) contains the output panel, which is displayed after the solver is initiated. The panel is divided into five tabs: Result, pH and EC, Solution, Anions and Cations.

3.5.3.1 Result

The Result tab summarises the optimisation output by comparing the calculated nutrient composition with the user-defined target composition. For each nutrient, the table reports the values specified in the Water vector, the Target vector and the values calculated by the solver in Achieved. In addition, a Final value is reported, representing the sum of the Water contribution and the Achieved contribution. The tab therefore links solver output to water contribution, target, achieved and final composition, together with deviation metrics. Concentrations are displayed in $\mu mol\,L^{-1}$ for values less than or equal to 0.1 mmol L^{-1} and in mmol L^{-1} for values above this threshold.

Deviation from the target composition is quantified using the error metrics `Abs. Error` and `Pct. Error`, representing the absolute deviation and the relative deviation in percent of `Final` from `Target`, respectively. To evaluate the overall goodness of fit across all nutrients, deviations are additionally aggregated as squared error terms. Absolute deviations are summarised as the sum of squared errors (`SSE`), whereas relative deviations are summarised as the sum of squared relative errors (`SSRE`).

3.5.3.2 pH & EC

The `pH & EC` tab summarises the calculated pH and EC values based on the `Final` composition. Calculations are performed for 25 °C using Davies activity correction, while complex formation is not considered. On the left side of the tab, the calculated `pH`, ionic strength mmol L^{-1} and total charge equivalents meq L^{-1} are reported. By selecting `Show pH charge breakdown`, an expanded view of the pH charge balance can be displayed, showing the contributions of fixed cations, fixed anions and variable components in meq L^{-1}.

On the right side of the tab, EC is reported in both mS cm^{-1} and µS cm^{-1}. These outputs apply the calculation procedures defined in the pH and EC methods sections to the final solution composition. From the calculated EC, osmotic potential is additionally derived and reported in `kPa`. By selecting `Show EC contributions`, the individual ionic contributions to the calculated EC can be displayed. For each ion, the table reports the concentration `c` (`mM`), charge number `z` and the contribution to conductivity mS cm^{-1}.

Above the `pH` and `EC` output sections, an `Adjust concentration` slider is provided with a range from `-100%` to `+100%`. This control scales the concentration of the reported `Final` composition and thereby allows interactive assessment of the resulting changes in pH, EC and related derived parameters.

3.5.3.3 Solution

The `Solution` tab reports the compounds used to obtain the `Achieved` composition. Solver quantities are translated into practical fertilizer amounts and stock-solution fractions. For each salt, acid, or base, the table reports the applied amount in mmol L^{-1}, the calculated molar mass in g mol^{-1} and the corresponding concentration in mg L^{-1}.

Above the `Fertilizer amounts` table, input fields are provided for working volume, stock factor and stock volume. Based on these inputs, additional columns are computed for the required amount for the selected working-solution volume (`x L`) and, when valid stock parameters are provided, for the selected stock preparation (`y L` of `z`-fold stock).

Compounds are grouped according to the BAK assignment logic into `A-BAK`, `B-BAK` and `Micro`. Assignment is rule-based: compounds containing micronutrients are assigned to `Micro`; compounds containing Ca, but no micronutrients, are assigned to A-BAK; compounds containing P or S, but neither micronutrients nor Ca, are assigned to B-BAK. Remaining compounds (AB) are assigned dynamically to A-BAK or B-BAK in order to balance the overall nutrient load between both tanks.

3.5.3.4 Anions and Cations

The Anions and Cations tabs report the relative proportions of selected anionic or cationic nutrient components on a mmol L^{-1} basis using ternary diagrams and provide a charge-based summary view of the final ionic composition. In the Anions tab, the default components are NO3-N, P and S, whereas in the Cations tab the default components are K, Ca and Mg. Alternative ions can also be selected for comparison, including Cl for anions and NH4-N or Na for cations. The respective proportions of Achieved and Target are displayed directly in the ternary graph and the corresponding percentage values are reported in the legend.

3.6 General recipe reconstruction and benchmark design

To evaluate solver performance, published crop-specific nutrient targets from Sonneveld & Straver (1994) were used as model input. These recipes specify target nutrient concentrations for different horticultural crops in mmol L^{-1}. For this evaluation, the solver was applied to each recipe using the salt stage only, without acids or bases. The resulting fertilizer compositions and achieved nutrient concentrations were compared with the specified target values using SSE and SSRE as error metrics.

Based on the achieved solution compositions, pH and EC were then calculated as derived physico-chemical output variables. These calculations were not used as optimisation targets in the recipe reconstruction, but served to evaluate the additional pH and EC reporting functions of the workflow. Because Sonneveld et al. (1999) and Sonneveld (2009) used one reference formulation to demonstrate a universal nutrient solution algorithm, the corresponding NutriCalc solution was compared in detail with the published formulation. This comparison was used as a benchmark against an established formulation approach.

3.7 Case study design

Three case studies were selected to demonstrate the application of the framework across a range of practical formulation scenarios. For this purpose, three user types with different formulation objectives were defined.

3.7.1 Case Study I: Alternative Hoagland II nutrient solution formulation

In this scenario, a plant scientist needs to prepare a classical Hoagland nutrient solution for an experiment using deionised water. The intended formulation is the macronutrient solution of the "Solution No. 2" according to Hoagland & Arnon (1938), consisting of 6 mM KNO_3, 4 mM $Ca(NO_3)_2$, 1 mM $NH_4H_2PO_4$ and 2 mM $MgSO_4 \cdot 7H_2O$. The scientist requires the solution at several concentration levels corresponding to target EC values of 0.5, 1.0, 1.5, 2.0, 2.5 and 3.0 mS cm^{-1}. However, the standard calcium nitrate salt is unavailable and only calcium ammonium nitrate (CAN; $5Ca(NO_3)_2 \cdot NH_4NO_3 \cdot 10H_2O$) is available as a substitute.

3.7.2 Case Study II: Tomato nutrient solution using Berlin-Dahlem water with acid adjustment

In this scenario, a horticultural technician needs to prepare a nutrient solution for tomato cultivation in rockwool using technical-grade fertilizers and tap water from Berlin-Dahlem (postal code 14195) instead of deionised water. The reference formulation is the tomato recipe reported by Sonneveld & Straver (1994). Because the tap water contributes background ions and alkalinity, the two-stage solver is applied with nitric acid permitted in the second stage for acid adjustment. Calculations are performed for the preparation of 500 L nutrient solution and for 18 L of a 50-fold concentrated stock solution separated into A-BAK, B-BAK and micronutrients.

3.7.3 Case Study III: Nutrient solution with increased sodium concentration for halophytes using mixed water sources

In this scenario, a horticultural scientist needs to formulate a nutrient solution for an experiment on sodium effects in *Salicornia europaea* using tap water from Berlin postal code 10997. Sodium treatments in the literature are frequently imposed using NaCl, but chloride has to be limited to 1.5 mmol L^{-1}. As the chloride concentration of the tap water exceeds this threshold, the tap water is mixed with demineralised water. The base recipe "Propagation vegetable plants in rockwool" reported by Sonneveld & Straver (1994) is used as the reference formulation. Ammonium nitrate (NH_4NO_3) is excluded from the set of permissible fertilizers. All calculations are performed for the preparation of 5 L stock solution.

4 Results

The results are presented in two main parts. The first part evaluates the model performance and plausibility of NutriCalc using crop-specific recipe reconstruction, selected reference-system checks for pH and EC calculations and comparison with a published formulation by Sonneveld & Straver (1994). The second part presents three case studies that demonstrate the application of NutriCalc to specific formulation problems and illustrate its practical applicability across different scenarios.

The evaluation logic was structured into four complementary layers. Recipe reconstruction was used to assess whether published crop-specific target vectors could be reproduced from the implemented fertilizer matrix under non-negativity constraints. Reference-system checks were used to evaluate the pH and EC modules under controlled chemical conditions, using phosphate-buffer data for pH and KCl conductivity data for EC. Comparison with the Sonneveld formulation served as an external benchmark against an established nutrient solution calculation procedure. Finally, the case studies were used as application demonstrations to examine workflow behaviour under practical constraints such as restricted fertilizer availability, source-water correction, acid adjustment and mixed-water formulation. They are therefore not interpreted as independent biological validation, but as demonstrations of applied use.

4.1 General evaluation across crop-specific recipes

The crop-specific recipe reconstruction evaluated whether the implemented fertilizer matrix and salt-stage NNLS solver could reproduce published horticultural target compositions from Sonneveld & Straver (1994).

4.1.1 Achieved nutrient solution compositions

All recipes listed in Table 3.1 were solved and the achieved formulation results are summarised in Table 4.1. Squared absolute and relative errors are reported for each recipe, quantifying the deviation between achieved and target nutrient concentrations across all nutrients. All achieved results matched the target nutrient profiles with SSE and SSRE values close to zero. Recipes including Si as a target nutrient exhibited comparatively higher relative error values. In formulations using Fe, achieved Na concentrations were non-zero although Na was not specified in the literature target vectors and Na contributed to the absolute error. When the unavoidable Na contribution associated with the selected Fe source was included in the target vector, the corresponding SSE values decreased to zero. For recipes containing Si, applying the second solver approach with reduced weighting for NO_3–N resulted in solutions with an SSRE of 0.

Table 4.1: Achieved crop-specific nutrient compositions with SSE and SSRE. Asterisks denote two-stage optimisation; parentheses indicate salt-only solver errors.

		mmol L⁻¹							µmol L⁻¹									
Category	Recipe	NO_3–N	NH_4–N	P	K	Ca	Mg	S	Na	Fe	Mn	Zn	B	Cu	Mo	Si	Squared absolute error	Squared relative error
Olericulture	Endive in Recirculating Water	19	1.25	2	9	5	1.5	1.13	39.94	39.94	5	4	30	0.75	0.5	0	**0.0016**	**0**
	Eggplant in Rockwool	15.5	1.5	1.25	6.75	3.25	2.5	1.5	15	15	10	5	30	0.75	0.5	0	**0.0002**	**0**
	Eggplant in Rockwool (RDW)	11.75	1	1	6.5	2.25	1.5	1.12	15	15	10	5	20	0.75	0.5	0	**0.0002**	**0**
	Bean in Rockwool	12	1	1.25	5.5	3.25	1.25	1.12	10	10	10	4	20	0.5	0.5	0	**0.0001**	**0**
	Courgette in Rockwool	16	1.25	1.25	7.25	3.62	2	1.25	10	10	10	5	30	0.75	0.5	0	**0.0001**	**0**
	Cucumber in Rockwool*	16	1.25	1.25	8	4	1.38	1.37	15	15	10	5	25	0.75	0.5	750	**0.0002 (0.9325)**	**0 (0.0053)**
	Cucumber in Rockwool (RDW)*	11.75	1	1.25	6.5	2.75	1	1	15	15	10	5	25	0.75	0.5	750	**0.0002 (0.9122)**	**0 (0.0097)**
	Propagation vegetable plants in rockwool	16.75	1.25	1.25	6.75	4.5	3	2.5	24.98	24.98	10	5	35	1	0.5	0	**0.0006**	**0**
	Sweet Pepper in Rockwool	15.5	1.25	1.25	6.5	4.75	1.5	1.75	15	15	10	5	30	0.75	0.5	0	**0.0002**	**0**
	Sweet Pepper in Rockwool (RDW)	12.75	1.25	1	5.75	3.25	1.13	1	15	15	10	4	25	0.75	0.5	0	**0.0002**	**0**
	Lettuce in Recirculating Water*	19	1.25	2	11	4.5	1	1.12	39.94	39.94	5	4	30	0.75	0.5	500	**0.0016 (0.423)**	**0 (0.0017)**
	Tomato in Rockwool	13.75	1.25	1.25	8.75	4.25	2	3.75	15	15	10	5	30	0.75	0.5	0	**0.0002**	**0**
	Tomato in Rockwool (RDW)	10.75	1	1.25	6.5	2.75	1	1.5	15	15	10	4	20	0.75	0.5	0	**0.0002**	**0**
Floriculture	Alstroemeria in Rockwool	11.25	1.25	1.25	6	2.87	1	1.25	24.98	24.98	10	4	25	0.75	0.5	0	**0.0006**	**0**
	Alstroemeria in Rockwool (RDW)	7.5	0.75	1	4.75	2	0.75	1.25	24.98	24.98	5	4	20	0.75	0.5	0	**0.0006**	**0**
	Anemone in Rockwool	13	1	1.5	6.5	3.75	1	1.25	34.96	34.96	5	4	30	0.75	0.5	0	**0.0012**	**0**
	Carnation in Rockwool or Peat	13	1	1.25	6.25	3.75	1	1.25	24.98	24.98	10	4	30	0.75	0.5	0	**0.0006**	**0**
	Carnation in Rockwool (reuse drainage)	7	0.75	0.8	4	1.62	0.6	0.7	19.99	19.99	5	3	20	0.5	0.5	0	**0.0004**	**0**
	Anthurium andreanum in Rockwool or Peat	4.5	0.8	0.7	3	1	0.7	1	15	15	3	3	20	0.5	0.5	0	**0.0002**	**0**
	Aster in Rockwool or Peat	13	1	1.25	6.25	3.75	1	1.25	24.98	24.98	10	4	30	0.75	0.5	0	**0.0006**	**0**
	Bouvardia in Rockwool	13	1.25	1.75	6	4.25	1	1.5	24.98	24.98	5	3.5	20	0.75	0.5	0	**0.0006**	**0**
	Bouvardia in Rockwool (RDW)	8	1	1.5	4	2.5	0.5	0.75	24.98	24.98	5	3.5	20	0.75	0.5	0	**0.0006**	**0**
	Chrysanthemum in Recirculating Water	12.75	1.25	1	7.5	2.5	1	1	59.78	59.78	20	3	20	0.5	0.5	0	**0.0036**	**0**
	Cymbidium in phenol foam	4.5	0.5	0.8	3	1.2	0.75	1.05	8	8	10	4	20	0.4	0.4	0	**0.0001**	**0**
	Cymbidium in rockwool/urethane foam	4	1	0.8	2.8	1	0.75	1.25	8	8	10	4	20	0.4	0.4	0	**0.0001**	**0**
	Euphorbia fulgens in Rockwool	11.5	1	1.5	6	3.5	1	1.5	34.96	34.96	10	3	20	0.5	0.5	0	**0.0012**	**0**
	Freesia in Rockwool or Sand	14.5	1.25	1.25	7.75	3.37	1.5	1.5	24.98	24.98	10	4	25	0.75	0.5	0	**0.0006**	**0**
	Gerbera in Rockwool	11.25	1.5	1.25	5.5	3	1	1.25	34.96	34.96	5	4	30	0.75	0.5	0	**0.0012**	**0**
	Gerbera in Rockwool (RDW)	7.25	0.75	0.6	4.5	1.6	0.4	0.7	24.98	24.98	5	3	20	0.5	0.5	0	**0.0006**	**0**
	Gypsophila in Rockwool	17	1.25	1.25	4	6	1.7	1.2	24.98	24.98	10	4	30	0.8	0.5	0	**0.0006**	**0**
	Hippeastrum in Pumice	13	1	1.25	7.5	3.12	1	1.25	10	10	10	5	30	0.5	0.5	0	**0.0001**	**0**
	Pot Plants in Expanded Clay	10.6	1.1	1.5	5.5	3	0.75	1	19.99	19.99	10	3	20	0.5	0.5	0	**0.0004**	**0**
	Rose in Rockwool	11	1.5	1.25	4.5	3.25	1.12	1.25	24.98	24.98	5	3.5	20	0.75	0.5	0	**0.0006**	**0**
	Rose in Rockwool (RDW)	4.3	0.85	0.5	2.15	0.9	0.5	0.5	15	15	5	3	15	0.5	0.5	0	**0.0002**	**0**
	Statice in Rockwool	12	1	1	6	3	1	1	15	15	10	5	2.5	0.75	0.5	0	**0.0002**	**0**
Fruticulture	Strawberry in Peaty Substrates	11.5	1	1	5.5	3.25	1.25	1.5	19.99	19.99	10	7	25	0.75	0.5	0	**0.0004**	**0**
	Strawberry in Recirculating Water	10	0.5	1.25	5.25	2.75	1.12	1.13	19.99	19.99	10	4	20	0.75	0.5	0	**0.0004**	**0**
	Melon in Rockwool*	16.25	1	1.25	7.5	4.75	1.25	1.5	10	10	10	4	20	0.5	0.5	750	**0.0001 (0.8753)**	**0 (0.005)**

4.1.2 Comparison with published formulation algorithm

The nutrient solution algorithm described by Sonneveld (2009) uses the cucumber recipe with reuse of drainage water published in Sonneveld & Straver (1994) as a demonstration of a universal nutrient solution formulation method proposed in Sonneveld et al. (1999). The target macronutrient concentrations for this demonstration are 11.75 mmol L^{-1} NO_3–N, 1.00 mmol L^{-1} NH_4–N, 1.25 mmol L^{-1} P, 6.50 mmol L^{-1} K, 2.75 mmol L^{-1} Ca, 1.00 mmol L^{-1} Mg and 1.00 mmol L^{-1} S. Micronutrients are not included because Sonneveld (2009) reports fertilizer additions only for macronutrients in this example. For a direct comparison, NutriCalc was constrained to the same set of salts reported by Sonneveld (2009). The published Sonneveld formulation includes nitric acid. NutriCalc matched the targets exactly (SSE = 0; SSRE = 0) using the one-stage solver without requiring acid addition. The same error metrics (SSE and SSRE) were computed for both the published Sonneveld formulation and the NutriCalc solution and are shown in Table 4.2. For the composition of nutrient salts used for the two formulations, see Table 4.3.

Table 4.2: Comparison of the published Sonneveld formulation and the corresponding NutriCalc reconstruction.

		Sonneveld solution			NutriCalc solution		
Nutrient	Target	Achieved	SE	SRE	Achieved	SE	SRE
NO_3–N	11.75	14.96	10.3041	0.0746	11.75	0.0000	0.0000
NH_4–N	1.00	1.25	0.0625	0.0625	1.00	0.0000	0.0000
P	1.25	1.50	0.0625	0.0400	1.25	0.0000	0.0000
K	6.50	8.88	5.6644	0.1341	6.50	0.0000	0.0000
Ca	2.75	2.47	0.0784	0.0104	2.75	0.0000	0.0000
Mg	1.00	1.07	0.0049	0.0049	1.00	0.0000	0.0000
S	1.00	0.87	0.0169	0.0169	1.00	0.0000	0.0000
			SSE = 16.1937	SSRE = 0.3434		SSE = 0.0000	SSRE = 0.0000

Table 4.3: Comparison of fertilizer additions in the published Sonneveld formulation and the corresponding NutriCalc reconstruction. Values are given in mmol L^{-1}.

		mmol L^{-1}	
Name	Formula	Sonneveld	NutriCalc
Calcium nitrate–ammonium nitrate decahydrate (5:1)	$5\ Ca(NO_3)_2 \cdot NH_4NO_3 \cdot 10\ H_2O$	0.49	0.55
Ammonium nitrate	NH_4NO_3	0.76	0.45
Monopotassium phosphate	KH_2PO_4	1.50	1.25
Magnesium nitrate hexahydrate	$Mg(NO_3)_2 \cdot 6\ H_2O$	1.07	1.00
Potassium sulfate	K_2SO_4	0.87	1.00
Nitric acid	HNO_3	1.00	–
Potassium nitrate	KNO_3	5.64	3.25

4.2 pH and EC outputs

For the crop-specific formulations, pH, ionic strength, total charge equivalents, electrical conductivity and osmotic potential were derived from the solver-derived salt compositions. The calculations were performed under salts-only conditions, with no background water contribution, using the Davies activity-correction model. Table 4.4 summarises the resulting values for the evaluated crop-specific formulations.

Table 4.4: Derived pH, ionic strength (I), total charge equivalents (TCE), electrical conductivity (EC) and osmotic potential for crop-specific formulations using Davies activity correction (salts-only; water = 0; KS 4.3 = 0; CO2(aq) = 0).

Category	Recipe	pH	I (mmol L^{-1})	TCE (meq L^{-1})	EC (mS cm^{-1})	Osmotic potential (kPa)
Olericulture	Endive in Recirculating Water	5.16	31.46	23.44	2.69	-96.98
	Eggplant in Rockwool	5.47	27.28	19.85	2.29	-82.51
	Eggplant in Rockwool (RDW)	5.57	20.15	15.09	1.8	-64.78
	Bean in Rockwool	5.45	21.34	15.57	1.82	-65.64
	Courgette in Rockwool	5.47	26.84	19.83	2.31	-82.99
	Cucumber in Rockwool	8.64	28.61	20.37	2.36	-85.02
	Cucumber in Rockwool (RDW)	8.72	21.61	15.4	1.81	-65.23
	Propagation vegetable plants in rockwool	5.46	33.41	23.14	2.6	-93.68
	Sweet Pepper in Rockwool	5.46	28.53	20.35	2.34	-84.19
	Sweet Pepper in Rockwool (RDW)	5.55	21.4	15.84	1.87	-67.4
	Lettuce in Recirculating Water	6.99	31.83	23.78	2.77	-99.67
	Tomato in Rockwool	5.46	32.78	22.6	2.59	-93.28
	Tomato in Rockwool (RDW)	5.46	20.53	15.09	1.79	-64.55
Floriculture	Alstroemeria in Rockwool	5.46	20.53	15.13	1.79	-64.47
	Alstroemeria in Rockwool (RDW)	5.43	15.38	11.12	1.33	-48.01
	Anemone in Rockwool	5.27	23.52	17.17	2	-72.01
	Carnation in Rockwool or Peat	5.46	23.16	16.88	1.98	-71.1
	Carnation in Rockwool (reuse drainage)	5.47	12.43	9.3	1.13	-40.74
	Anthurium andreanum in Rockwool or Peat	5.45	10.13	7.28	0.89	-32.04
	Aster in Rockwool or Peat	5.46	23.16	16.88	1.98	-71.1
	Bouvardia in Rockwool	5.2	24.89	17.88	2.06	-74.19
	Bouvardia in Rockwool (RDW)	5.27	15.13	11.12	1.31	-47.26
	Chrysanthemum in Recirculating Water	5.73	21.15	16.04	1.92	-69.12
	Cymbidium in phenol foam	5.67	10.58	7.46	0.9	-32.47
	Cymbidium in rockwool/urethane foam	5.67	10.48	7.36	0.9	-32.27
	Euphorbia fulgens in Rockwool	5.36	22.53	16.17	1.88	-67.72
	Freesia in Rockwool or Sand	5.45	25.53	18.88	2.22	-79.81
	Gerbera in Rockwool	5.33	20.76	15.16	1.79	-64.29
	Gerbera in Rockwool (RDW)	5.58	12.32	9.37	1.16	-41.65
	Gypsophila in Rockwool	5.44	29.96	20.78	2.33	-83.74
	Hippeastrum in Pumice	5.47	22.34	16.82	2	-72.09
	Pot Plants in Expanded Clay	5.37	19.19	14.21	1.68	-60.36
	Rose in Rockwool	5.32	20.76	14.87	1.73	-62.43
	Rose in Rockwool (RDW)	5.67	7.94	5.88	0.73	-26.15
	Statice in Rockwool	5.57	20.28	15.09	1.79	-64.59
Fruticulture	Strawberry in Peaty Substrates	5.62	21.85	15.62	1.83	-65.84
	Strawberry in Recirculating Water	5.47	18.84	13.61	1.6	-57.48
	Melon in Rockwool	8.73	29.87	20.89	2.4	-86.29

4.2.1 Performance of pH calculation

Both γ activity correction models (Davies and Debye–Hückel) were evaluated for pH prediction against the phosphate buffer system (50 mL of 0.1 mol L^{-1} KH_2PO_4 + x mL of 0.1 mol L^{-1} NaOH), which spans pH 5.8–8.0 (Bower & Bates, 1955; Haynes, 2017). In Figure 4.1, predicted pH values are plotted against the reference pH for increasing NaOH additions at 25 °C. The Davies model closely matched the reference values (R^2 = 0.998), whereas the Debye–Hückel model showed weaker agreement (R^2 = 0.880).

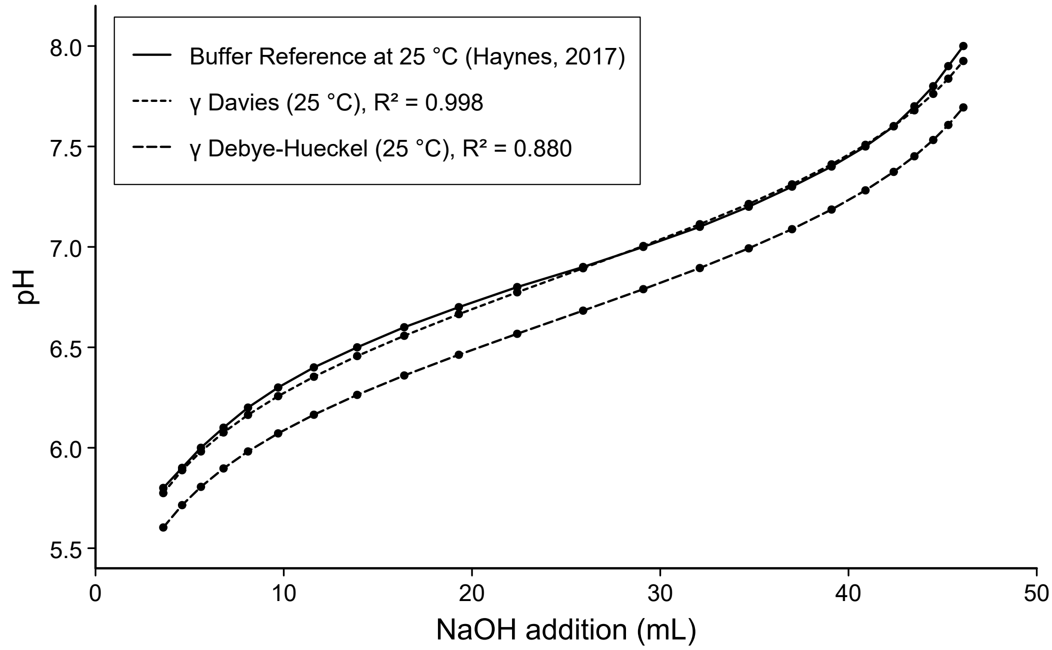


Figure 4.1: Comparison of calculated and reference pH values for a phosphate buffer prepared from 50 mL of 0.1 mol L^{-1} KH_2PO_4 and increasing additions of 0.1 mol L^{-1} NaOH at 25 °C.

4.2.2 Performance of EC calculation

To evaluate EC calculation performance, model predictions were compared against reference electrical-conductivity values for KCl solutions at 25 °C. Electrical-conductivity values for 0.01 and 0.1 mol L^{-1} KCl were taken from Haynes (2017), which reports values based on primary conductivity standards originally published by Pratt et al. (2001). In Table 4.5, these reference values are compared with NutriCalc EC predictions using the Davies and Debye–Hückel activity-coefficient models as reported by Kalka (2020). For comparison, the empirical approach by Sonneveld et al. (1999) and geochemical computations with PHREEQC (Parkhurst & Appelo, 2013) are also shown. PHREEQC and NutriCalc with γ Davies were very close to the values reported by Haynes (2017) at both concentrations. NutriCalc with γ Debye–Hückel slightly underestimated EC, while the underestimation was larger for the Sonneveld empirical approach.

Table 4.5: Comparison of reference and calculated electrical conductivity values for KCl solutions at 25 °C.

	µS cm⁻¹				
KCl (mol L^{-1})	CRC	Sonneveld	NutriCalc Debye–Hückel	NutriCalc Davies	PHREEQC
0.01	1408	1140	1396	1408	1409
0.10	12825	9690	11996	12923	12988

The comparison against the EC values reported by Sonneveld & Straver (1994) showed distinct model-specific deviation patterns as shown in Figure 4.2. The Sonneveld empirical approximation produced the smallest typical deviation but showed a slight negative bias overall (ME = −0.03 mS cm^{-1}, MAE = 0.06 mS cm^{-1}). Its largest signed underestimation was −0.18 mS cm^{-1}, while the maximum overestimation reached +0.15 mS cm^{-1}.

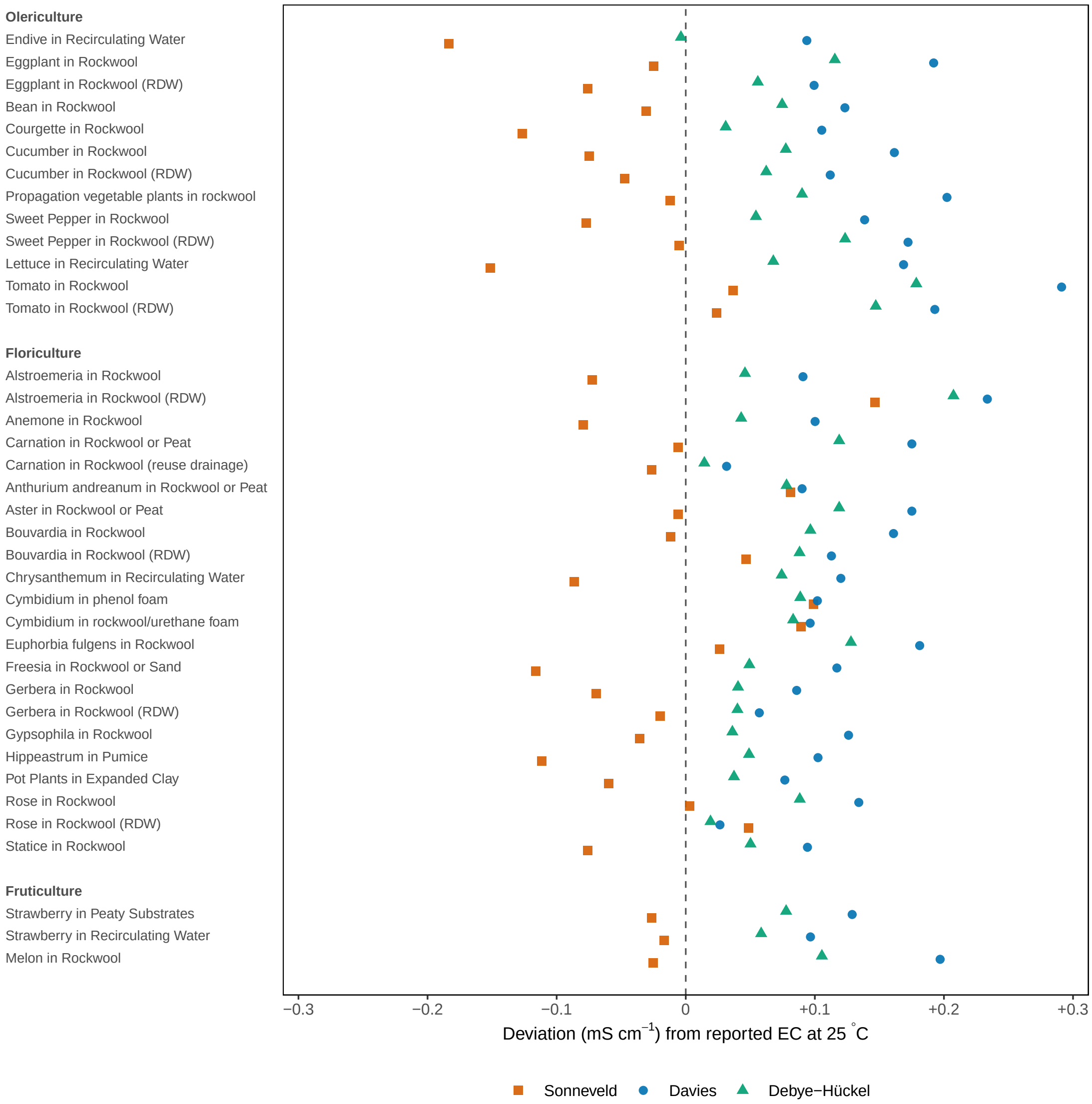


Figure 4.2: Deviations between calculated and reported electrical conductivity for crop-specific nutrient solutions from Sonneveld & Straver (1994). Deviation = calculated − reported (mS cm^{-1}). Calculated values are shown for three approaches: the Sonneveld (1999) empirical EC estimate, NutriCalc with Davies activity correction and NutriCalc with Debye–Hückel activity correction. Positive values indicate overestimation; negative values indicate underestimation.

Both NutriCalc EC models tended to overestimate the reported values. The Debye–Hückel model showed a smaller systematic bias and typical error size (ME = +0.08 mS cm^{-1}, MAE = 0.08 mS cm^{-1}) than the Davies model (ME = +0.13 mS cm^{-1}, MAE = 0.13 mS cm^{-1}). The maximum overestimation was +0.21 mS cm^{-1} for Debye–Hückel and +0.29 mS cm^{-1} for Davies across all 38 recipes. Thus, the empirical Sonneveld approximation reproduced the reported EC values most closely in absolute terms, whereas the EC models showed a small positive bias under the salts-only reconstruction assumptions. EC values and deviations for all models are shown in the Appendix under Table A.1.

4.3 Case Study Results

4.3.1 Case study I — Experimental scientist: classical solution reconstruction under restricted salt availability (deionised water) and different EC.

The macronutrient composition of Hoagland solution No. 2 consists of 14 mmol L^{-1} NO_3–N, 1 mmol L^{-1} NH_4–N, 1 mmol L^{-1} P, 6 mmol L^{-1} K, 4 mmol L^{-1} Ca, 2 mmol L^{-1} Mg and 2 mmol L^{-1} S. In the absence of $Ca(NO_3)_2$ three exemplary alternative formulations using the technical salt CAN are shown in Table 4.6.

Table 4.6: Original Hoagland solution No. 2 fertilizer combination and three alternative fertilizer combinations achieving the same target macronutrient composition. Values are given in mmol L^{-1}.

Salt	Original	Alternative 1	Alternative 2	Alternative 3
$Ca(NO_3)_2$	4.00	—	—	—
$5Ca(NO_3)_2{\cdot}NH_4NO_3{\cdot}10H_2O$	—	0.80	0.80	0.80
KNO_3	6.00	1.20	1.00	1.20
$MgSO_4{\cdot}7H_2O$	2.00	—	—	—
$Mg(NO_3)_2{\cdot}6H_2O$	—	2.00	2.00	2.00
$NH_4H_2PO_4$	1.00	—	—	0.20
KH_2PO_4	—	1.00	1.00	0.80
K_2SO_4	—	1.90	2.00	2.00
NH_4NO_3	—	—	0.20	—
$(NH_4)_2SO_4$	—	0.10	—	—
Number of salts	**4**	**6**	**6**	**6**

4.3.1.1 Achieved composition

The target macronutrient composition of Hoagland solution No. 2 was achieved using 0.80 mmol L^{-1} $5Ca(NO_3)_2{\cdot}NH_4NO_3{\cdot}10H_2O$, 0.80 mmol L^{-1} KH_2PO_4, 5.20 mmol L^{-1} KNO_3, 2.00 mmol L^{-1} $MgSO_4{\cdot}7H_2O$ and 0.20 mmol L^{-1} $NH_4H_2PO_4$. This is shown as Alternative 3 in Table 4.6. Values for the different EC designs are shown in Table 4.7.

Table 4.7: Fertilizer salt amounts required to prepare modified Hoagland solution No. 2 at target EC values from 0.5 to 3.0 mS cm⁻¹. Values are given in mmol L⁻¹.

	Concentration in mmol L^{-1} for target EC (mS cm^{-1})					
Salt / parameter	EC 0.5	EC 1.0	EC 1.5	EC 2.0	EC 2.5	EC 3.0
Strength ($5Ca(NO_3)_2 \cdot NH_4NO_3 \cdot 10H_2O$	0.17	0.34	0.54	0.73	0.92	1.12
KH_2PO_4	0.17	0.34	0.54	0.73	0.92	1.12
KNO_3	1.09	2.24	3.48	4.73	5.98	7.28
$MgSO_4 \cdot 7H_2O$	0.42	0.86	1.34	1.82	2.30	2.80
$NH_4H_2PO_4$	0.04	0.09	0.13	0.18	0.23	0.28

4.3.1.2 Deviations from target

No deviation from the target macronutrient composition of Hoagland solution No. 2 was observed in the achieved solutions. Both the original formulation and the alternative fertilizer combinations matched the target composition exactly, resulting in SSE and SSRE values of 0.00.

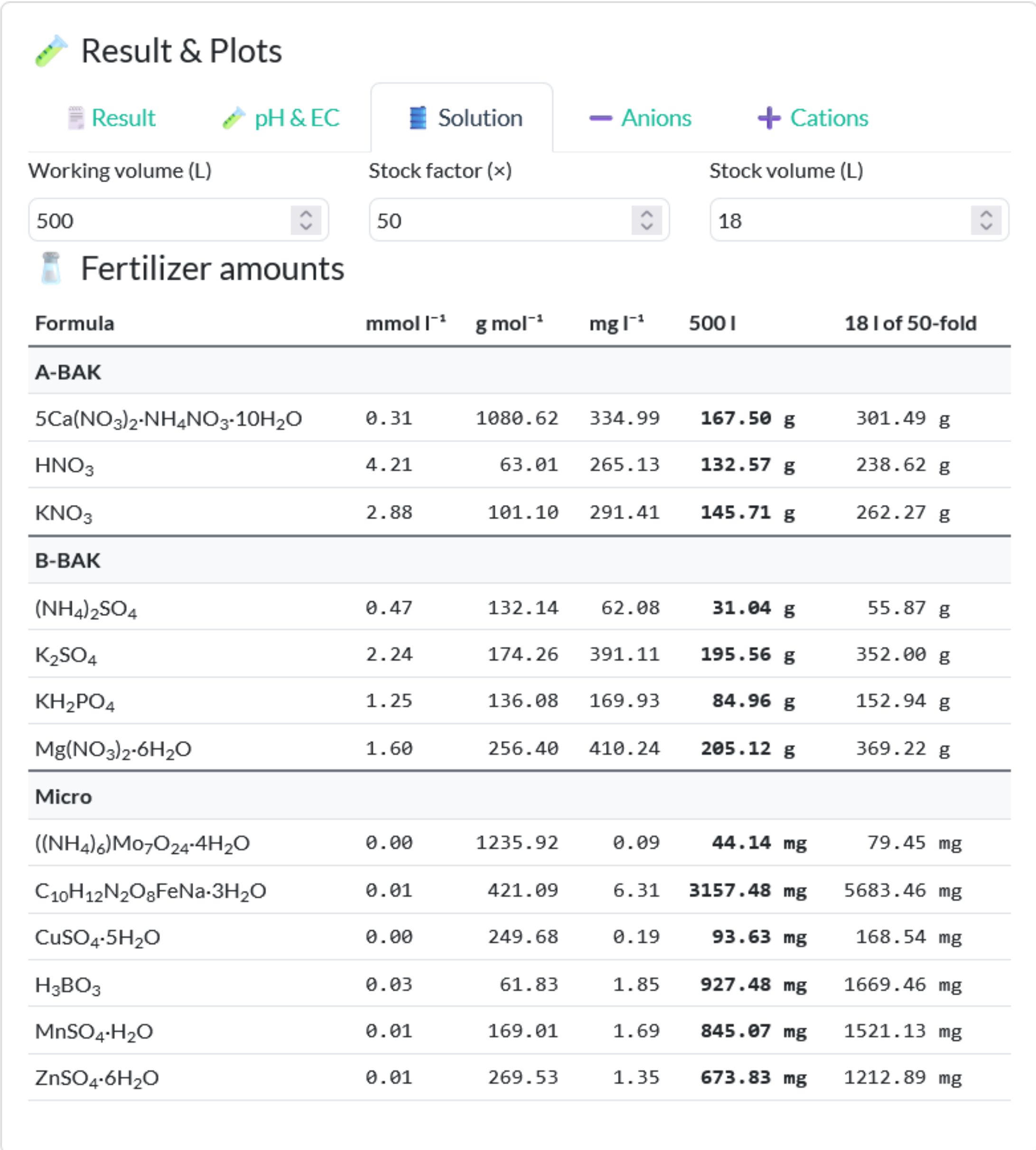
4.3.1.3 pH and EC

The calculated undiluted reconstructed Hoagland solution No. 2 had a pH of 5.08. Ionic strength amounted to 27.02 mmol L^{-1}, while the total concentration of charge equivalents was 19.01 meq L^{-1}. The calculated EC was 2.19 mS cm^{-1} and the osmotic potential was −78.82 kPa. The largest ionic contributions to EC were from NO_3^- (0.91 mS cm^{-1}), K^+ (0.40 mS cm^{-1}) and Ca^{2+} (0.37 mS cm^{-1}). The EC scaling series for the alternative formulation is reported separately in Table 4.7.

4.3.2 Case study II — Horticultural technician: tomato nutrient formulation using Berlin-Dahlem water with acid adjustment

4.3.2.1 Achieved composition

For a nutrient solution formulated for tomatoes cultivated in rockwool (Sonneveld & Straver, 1994) and prepared with tap water from postal code 14195 (BWB, 2026), the two-stage solver with nitric acid addition was used. The calculated fertilizer salts are shown in Figure 4.3, including the amounts required for the preparation of 500 L nutrient solution and the corresponding amounts for 18 L of a 50-fold concentrated stock solution separated into A-BAK, B-BAK and micronutrients.



Result & Plots

Result | pH & EC | Solution | Anions | Cations

Working volume (L): 500

Stock factor (×): 50

Stock volume (L): 18

Fertilizer amounts

Formula	mmol l⁻¹	g mol⁻¹	mg l⁻¹	500 l	18 l of 50-fold
A-BAK					
$5Ca(NO_3)_2{\cdot}NH_4NO_3{\cdot}10H_2O$	0.31	1080.62	334.99	**167.50** g	301.49 g
HNO_3	4.21	63.01	265.13	**132.57** g	238.62 g
KNO_3	2.88	101.10	291.41	**145.71** g	262.27 g
B-BAK					
$(NH_4)_2SO_4$	0.47	132.14	62.08	**31.04** g	55.87 g
K_2SO_4	2.24	174.26	391.11	**195.56** g	352.00 g
KH_2PO_4	1.25	136.08	169.93	**84.96** g	152.94 g
$Mg(NO_3)_2{\cdot}6H_2O$	1.60	256.40	410.24	**205.12** g	369.22 g
Micro					
$((NH_4)_6)Mo_7O_{24}{\cdot}4H_2O$	0.00	1235.92	0.09	**44.14** mg	79.45 mg
$C_{10}H_{12}N_2O_8FeNa{\cdot}3H_2O$	0.01	421.09	6.31	**3157.48** mg	5683.46 mg
$CuSO_4{\cdot}5H_2O$	0.00	249.68	0.19	**93.63** mg	168.54 mg
H_3BO_3	0.03	61.83	1.85	**927.48** mg	1669.46 mg
$MnSO_4{\cdot}H_2O$	0.01	169.01	1.69	**845.07** mg	1521.13 mg
$ZnSO_4{\cdot}6H_2O$	0.01	269.53	1.35	**673.83** mg	1212.89 mg

Figure 4.3: Fertilizer composition calculated in the NutriCalc Shiny app for a tomato nutrient solution in rockwool using tap water from Berlin postal code 14195. The formulation is separated into A-BAK, B-BAK and micronutrient stock solution fractions. Fertilizer requirements are given for the preparation of 500 L nutrient solution and for 18 L of 50-fold concentrated stock solution.

4.3.2.2 Deviations from target

The tomato recipe specifies a target composition without Na, Cl, or Si. In the calculated solution, these elements were present due to their contribution from the tap water. Although the target composition was matched with an SSRE of 0, absolute deviations remained for Na, Cl and Si, resulting in an SSE of 7.69. The absolute errors were 2.03 mmol L^{-1} for Na, 1.86 mmol L^{-1} for Cl and 0.30 mmol L^{-1} for Si. Of the Na concentration, 0.015 mmol L^{-1} originated from Fe-EDTA.

4.3.2.3 pH and EC

The tap water from postal code 14195 had a calculated pH of 7.44 and an EC of 0.86 mS cm^{-1}. Besides the ions considered in the target nutrient composition, it supplied 2.02 mmol L^{-1} Na^+ and 1.86 mmol L^{-1} Cl^-, contributing 0.09 and 0.13 mS cm^{-1}, respectively, to the total EC. The electroneutral 0.3 mmol L^{-1} Si present in the tap water did not contribute to the overall EC.

The calculated final nutrient solution had a pH of 5.82. Ionic strength amounted to 34.84 mmol L^{-1}, while the total concentration of charge equivalents was 24.61 meq L^{-1}. The calculated EC was 2.81 mS cm^{-1} and the osmotic potential was −101.02 kPa. In the final solution, the largest ionic contribution to EC was from NO_3^- at 0.89 mS cm^{-1}, followed by K^+ at 0.58 mS cm^{-1} and SO_4^{2-} at 0.45 mS cm^{-1}.

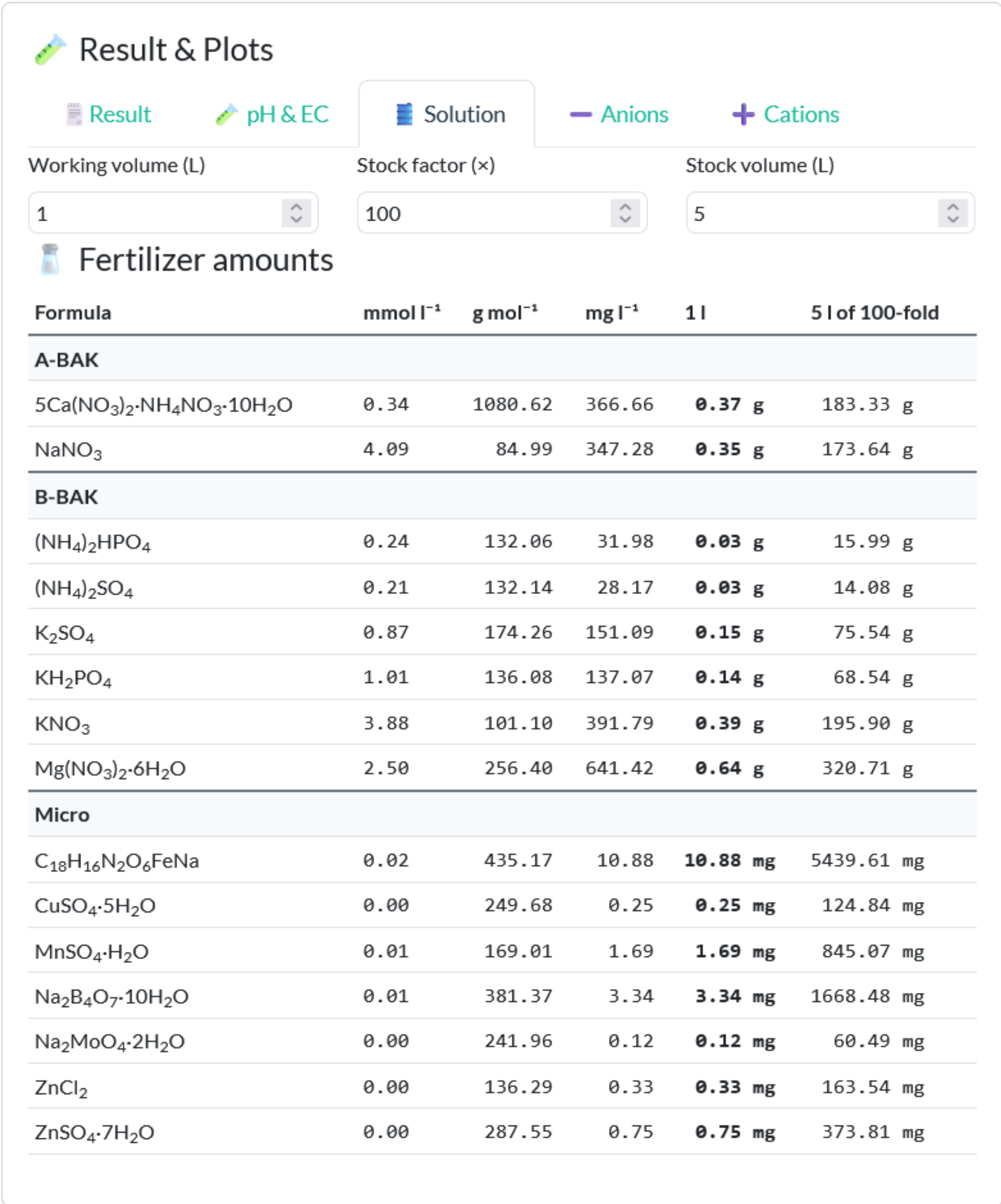
4.3.3 Case study III — Research scientist: nutrient formulation for a novel halophyte (Salicornia) using mixed water sources

4.3.3.1 Achieved composition

The calculated mixed water consists of 90 % tap water from Berlin postal code 10997 and 10 % demineralised water, thereby reducing the chloride concentration of the water source to 1.5 mmol L^{-1}. Based on this water composition and the reference recipe "Propagation vegetable plants in rockwool" reported by Sonneveld & Straver (1994), chloride is set to 1.5 mmol L^{-1}, yielding a calculated sodium concentration of 5.5 mmol L^{-1} to maintain electroneutrality. Since ammonium nitrate (NH_4NO_3) is excluded from the permissible fertilizer set, the target composition is achieved using an alternative fertilizer combination. The final fertilizer amounts are expressed for the preparation of 5 L stock solution. The calculated fertilizer composition is shown in Figure 4.4.

4.3.3.2 Deviations from target

Only negligible deviations from the target composition were observed in the achieved solution (SSE = 0.07; SSRE = 0). The calculated solution contained 5.5 mmol L^{-1} Na and 1.5 mmol L^{-1} Cl, consistent with the target composition and the mixed water source. The only remaining absolute deviation concerned Si contributed by the tap water, which was not specified as a target and therefore caused a small SSE.



Formula	mmol l^{-1}	g mol^{-1}	mg l^{-1}	1 l	5 l of 100-fold
A-BAK					
$5Ca(NO_3)_2 \cdot NH_4NO_3 \cdot 10H_2O$	0.34	1080.62	366.66	**0.37 g**	183.33 g
$NaNO_3$	4.09	84.99	347.28	**0.35 g**	173.64 g
B-BAK					
$(NH_4)_2HPO_4$	0.24	132.06	31.98	**0.03 g**	15.99 g
$(NH_4)_2SO_4$	0.21	132.14	28.17	**0.03 g**	14.08 g
K_2SO_4	0.87	174.26	151.09	**0.15 g**	75.54 g
KH_2PO_4	1.01	136.08	137.07	**0.14 g**	68.54 g
KNO_3	3.88	101.10	391.79	**0.39 g**	195.90 g
$Mg(NO_3)_2 \cdot 6H_2O$	2.50	256.40	641.42	**0.64 g**	320.71 g
Micro					
$C_{18}H_{16}N_2O_6FeNa$	0.02	435.17	10.88	**10.88 mg**	5439.61 mg
$CuSO_4 \cdot 5H_2O$	0.00	249.68	0.25	**0.25 mg**	124.84 mg
$MnSO_4 \cdot H_2O$	0.01	169.01	1.69	**1.69 mg**	845.07 mg
$Na_2B_4O_7 \cdot 10H_2O$	0.01	381.37	3.34	**3.34 mg**	1668.48 mg
$Na_2MoO_4 \cdot 2H_2O$	0.00	241.96	0.12	**0.12 mg**	60.49 mg
$ZnCl_2$	0.00	136.29	0.33	**0.33 mg**	163.54 mg
$ZnSO_4 \cdot 7H_2O$	0.00	287.55	0.75	**0.75 mg**	373.81 mg

Figure 4.4: Fertilizer composition calculated in the NutriCalc Shiny app for the halophyte case study using mixed source water. The output shows the translation of the solver result into A-BAK, B-BAK and micronutrient stock-solution fractions, including fertilizer requirements for 5 L of 100-fold stock solution.

4.3.3.3 pH and EC

The calculated final nutrient solution had a pH of 7.19. Ionic strength amounted to 39.47 mmol L^{-1}, while the total concentration of charge equivalents was 28.60 meq L^{-1}. The calculated EC was 3.09 mS cm^{-1} and the osmotic potential was −111.29 kPa. The largest ionic contributions to EC were from NO_3^- (1.07 mS cm^{-1}), K^+ (0.45 mS cm^{-1}) and Ca^{2+} (0.39 mS cm^{-1}). In addition, Na^+ and Cl^- contributed 0.25 and 0.10 mS cm^{-1}, respectively.

4.4 Summary of validation and application demonstration

Taken together, the preceding analyses combined model-performance checks, benchmark comparisons and application demonstrations. The crop-specific recipe reconstructions assessed whether feasible literature targets could be reproduced from the implemented fertilizer matrix. The comparison with the Sonneveld algorithm provided a benchmark against an established formulation approach. The phosphate-buffer and KCl comparisons evaluated the pH and EC modules against defined reference systems. As a secondary recipe-level comparison, calculated EC values were compared with the EC values reported for the crop-specific recipes by Sonneveld & Straver (1994) to assess whether the implemented EC models produced values in the expected practical range for horticultural nutrient solutions.

Finally, the three case studies demonstrated the application of NutriCalc under practical constraints involving restricted fertilizer availability, source-water correction, acid adjustment and mixed-water formulation. The case studies are therefore interpreted as application demonstrations rather than independent validation tests.

5 Discussion

5.1 Solver robustness and formulation performance

Across the evaluated formulations, the solver reproduced most published target compositions exactly or with only very small residual deviations. This indicates that the implemented fertilizer matrix is sufficiently expressive for a broad range of horticultural nutrient solutions. In particular, the crop-specific targets from Sonneveld & Straver (1994) were matched with SSE and SSRE values close to zero in most cases, showing that the combination of stoichiometric matrix construction and normalised NNLS optimisation is capable of representing many practically relevant formulation problems with high fidelity.

Where deviations remained, they were chemically interpretable rather than arbitrary numerical failures. This is important because it means that residual error in the present framework should not immediately be interpreted as an algorithmic weakness. Instead, it often reflects specific properties of the fertilizer representation or of the target vector itself.

Salts used in the solver were selected on the basis of their solubility and their reported use in hydroponics. Salts that are not water soluble but may still be relevant in soil based systems were therefore not included in the implemented salt matrix.

The small SSE values reported for the crop set of Sonneveld & Straver (1994) are mainly caused by the use of chelated Fe sources in the salt matrix. In the implemented representation, Fe chelates introduce Na as a counter ion. Accordingly, a non zero Na contribution can occur in the achieved solution even when Na is absent from the published target vector. This should not be interpreted as a substantive mismatch in the nutritionally relevant targets, but rather as a consequence of using chemically realistic Fe fertilizers. Consistent with this interpretation, SSE drops to zero when the Na target is set equal to the Fe associated Na contribution.

A second recurring case involves recipes containing Si, commonly at around 0.75 mmol L^{-1}. In the present matrix, Si is supplied via K_2SiO_3, which necessarily introduces additional K and also behaves as a strongly basic reagent. Consequently, exact matching depends on whether this coupled K contribution is represented in the target vector and or corrected by acid addition. When Si is omitted, or when the corresponding K contribution is accounted for explicitly, SSRE approaches zero.

This indicates that the residual mismatch in Si containing recipes is not primarily caused by numerical instability of the optimiser, but by chemically coupled fertilizer behaviour. In other words, the difficulty lies less in the NNLS procedure itself than in the fact that silicon supply through potassium silicate cannot be treated independently of its accompanying alkalinity and counter ion contribution, because it behaves effectively as a base used in the role of a salt.

Although Si is generally considered a beneficial rather than an essential element, positive effects on yield and disease suppression have been reported (Rengel et al., 2023; Voogt & Sonneveld, 2001). Practical use of potassium silicate often requires stoichiometric acid compensation. In the formulations discussed by Voogt & Sonneveld (2001), this was addressed by combining silicate addition with nitric acid and with corresponding adjustment of nitrate salts. The fact that Si containing cases improved when acid and base correction was allowed in NutriCalc therefore supports rather than undermines the chemical plausibility of the framework. Taken together, the results suggest that NutriCalc performs robustly within the tested horticultural domain and that the remaining deviations can usually be traced back to identifiable chemical causes.

The three case studies illustrate distinct formulation constraints. The Hoagland case showed that exact reconstruction remained possible even when the original calcium nitrate source was unavailable and replaced by calcium ammonium nitrate. The tomato case demonstrated the effect of source-water correction, where Na, Cl and Si from Berlin-Dahlem tap water remained present in the final solution. The halophyte case extended the workflow to mixed-water formulation, where sodium and chloride constraints were part of the experimental design. Together, these examples show that NutriCalc can be applied not only to recipe reconstruction, but also to constrained formulation scenarios involving fertilizer substitution, source-water chemistry and experimental manipulation of specific ions.

5.2 Exact solvability, electroneutrality and non-uniqueness

In this context, an electroneutral target is defined as a target vector for which the sum of cation and anion charge equivalents is balanced. A non electroneutral target is a target vector for which this condition is not fulfilled. Electroneutrality is a necessary but not sufficient condition for exact solvability when only electrically neutral salts are considered. Exact solvability also depends on the selected fertilizer set.

If the target is electroneutral and the available fertilizers span the required composition, the solver may find an exact solution. In such cases, both SSE and SSRE are zero, indicating that all targets have been matched exactly in both absolute and relative terms. However, this does not imply that the solution is unique. Different fertilizer combinations may produce the same final nutrient composition (Steiner, 1961). Some alternative solutions differ only in the hydration state of a salt, for example $Ca(NO_3)_2$ versus $Ca(NO_3)_2 \cdot 4H_2O$, whereas others may involve chemically distinct fertilizer combinations. In some cases, the number of exact solutions may be effectively infinite because different salts can compensate for one another while preserving the same final nutrient profile.

Accordingly, the NNLS solver returns one admissible non negative optimum, but it does not characterise the full set of possible exact solutions. Users can nevertheless influence the returned solution by restricting the fertilizer set through the selection and deselection of salts. In addition, small fertilizer amounts may appear in the solution if they provide even a minor reduction in the objective function. Thus, an exact or near exact solution is not necessarily the chemically simplest or most practical one.

If the target is non electroneutral, it cannot be matched exactly using only electrically neutral salts. In that case, the solver returns the non negative best fit solution that minimises the residual error. Because NutriCalc normalises residuals by target magnitude, the optimisation criterion primarily minimises the sum of squared relative errors for non zero targets and the sum of squared absolute errors for zero targets.

A useful implication of this formulation is that infeasible targets are not solved in an intuitively arithmetic sense, but according to the relative structure of the objective function. When a target cannot be matched exactly, the remaining mismatch is distributed according to the row scaling and the imposed weights. Thus, residual error should not be interpreted simply as failure. It may instead indicate that the target is not electroneutral, that the selected fertilizer set does not span the target composition, or that chemically realistic fertilizer sources introduce unavoidable side contributions.

5.3 Error metrics and interpretation of residual deviation

Reported errors quantify the deviation between target and achieved nutrient concentrations. For each nutrient, the formulation output reports the absolute error and, when the target is non zero, the relative error. Two aggregate metrics are derived: the sum of squared errors, SSE and the sum of squared relative errors, SSRE. The per nutrient errors indicate which nutrients deviate from the target, by how much and in which direction. The squared error metrics summarise overall solution quality.

The optimisation itself is based on a row normalised non negative least squares, NNLS, formulation. Residuals are scaled by a nutrient specific factor: for nutrients with non zero targets, deviations are normalised by the corresponding target concentration. For nutrients with zero targets, a scaling factor of one is applied. As a result, all nutrients, including those with targets set to zero, are actively included in the objective function. Any non zero achieved concentration for a zero target nutrient contributes a quadratic penalty and is therefore discouraged during optimisation.

SSE reflects the total squared deviation across all nutrients and therefore captures leakage into zero target nutrients. SSRE, in contrast, summarises squared relative deviations only for nutrients with non zero targets and serves as the primary performance indicator for how closely specified targets were matched. However, both SSE and SSRE are post hoc reporting metrics. The solver decision is determined by the normalised and weighted NNLS objective. Consequently, a formulation with slightly lower relative deviation for non zero targets may still be rejected if it introduces substantial unintended supply of zero target nutrients, because this increases the overall objective value.

Without normalisation, an objective based solely on squared absolute errors would bias the optimisation toward macronutrients, since deviations in the µmol L^{-1} range would be numerically negligible compared with deviations in the mmol L^{-1} range. Row wise normalisation therefore ensures that proportional deviations are penalised on a comparable scale, preventing magnitude driven bias and enabling balanced optimisation for both macro and micronutrients.

In this sense, SSE and SSRE should not be interpreted as direct optimisation targets, but as complementary descriptors of the achieved solution. SSE is particularly informative for identifying chemically relevant leakage into nutrients that were intended to remain absent, whereas SSRE is more informative for evaluating proportional agreement with specified non zero targets.

A useful illustration is the 1:1 salt coupling case, for example NaCl or NH_4NO_3. If the targets are 1 mmol L^{-1} for one ion and 2 mmol L^{-1} for the coupled counter ion, relative error minimisation yields an optimum at 1.2 mmol L^{-1} rather than the arithmetic midpoint of 1.5 mmol L^{-1}. This demonstrates a non intuitive but important property of the normalised objective: the optimum is shifted toward the smaller target because deviations are scaled relative to target magnitude.

5.4 Weighting and two-stage optimisation as practical control tools

Weighting is optional, so the solver can also be used in an unweighted form, in which all nutrients contribute equally after normalisation. If weights are applied, the problem becomes a weighted non negative least squares problem, in which deviations in highly weighted nutrients contribute more strongly to the objective function than deviations in nutrients with lower weights. Lowering the weight of one nutrient shifts more of the residual error to that nutrient and allows the remaining nutrients to be matched more closely. In practical terms, this means that one nutrient can be deliberately sacrificed to improve the overall fit of the others when the target profile is difficult or impossible to achieve with the selected salts.

This feature is particularly useful for chemically constrained formulations. If one nutrient cannot be matched simultaneously with all others because of stoichiometric coupling within the available fertilizers, reducing its weight may substantially improve the fit for the remaining targets. When the second solver stage is enabled, additional adjustment with acids or bases may further improve the final composition and partially compensate for such trade offs.

The one-stage results show that the two-stage solver is not required for most standard recipes, but becomes useful in chemically coupled cases. The two-stage solver nevertheless provides an important extension when an exact solution is not achievable with neutral salts alone. In this architecture, the first stage fits the composition using neutral fertilizer salts, whereas acids and bases are introduced only in a second corrective step. This separation is chemically meaningful because it prevents acids and bases from acting as unrestricted first stage degrees of freedom that could improve the numerical fit while reducing interpretability of the resulting fertilizer composition.

The results support this design particularly for pH active or stoichiometrically awkward cases, such as silicate containing formulations. The two-stage framework should therefore not be understood merely as a numerical convenience, but as a practically motivated control mechanism that balances solvability, chemical interpretability and operational realism.

5.5 Electroneutrality and pH as coupled constraints

Electroneutrality is one of the most fundamental principles in nutrient solution formulation. In any aqueous electrolyte solution, the sum of positive charges must equal the sum of negative charges. Thus, the total charge contributed by cations is balanced by the total charge contributed by anions (Kalka, 2024). While many dissolved compounds dissociate into well defined cations and anions and therefore contribute predictably to charge balance, some nutrients exist in different ionic species depending on pH. At the same time, pH should not be understood merely as a measure of hydrogen ion concentration, but more accurately as a measure of hydrogen ion activity. From this perspective, pH can be interpreted as a dependent variable constrained by the electroneutrality condition in aqueous solution (Stumm & Morgan, 1996). Accordingly, the activities of H_3O^+ and OH^- adjust in conjunction with pH dependent speciation systems such as phosphate, sulfate, carbonate, borate and silicate to maintain charge balance, rather than being determined solely by the stoichiometric addition and neutralisation of acid and base.

Electroneutrality ultimately determines whether a target nutrient composition is physically realisable. In the optimisation procedure, non negativity of salt addition was implemented explicitly as a mathematical constraint, whereas electroneutrality was not imposed as an additional constraint. This is because fertilizer salts contribute ions in stoichiometrically balanced combinations. Acid or base addition changes pH and ionic speciation without violating the overall charge balance of the solution. Consequently, if a target nutrient profile is itself not electroneutral, no exact solution exists and the solver can only minimise the deviation from the specified targets.

This is directly relevant for interpretation of pH within the framework. pH should not be treated as an independently selectable scalar, but as an emergent property of charge balance, acid base equilibria and activities. Integrating charge balance based pH estimation therefore extends formulation from simple nutrient matching toward chemically coherent solution evaluation.

5.6 Role of pH and EC modelling in the workflow

A notable strength of the framework is that formulation is not treated as complete once nutrient targets have been matched. Estimated pH and EC provide a second interpretive layer linking nutrient composition to broader physicochemical behaviour. This is relevant because two solutions with similar nominal nutrient targets can differ in buffering, ionic strength, conductivity or pH response depending on fertilizer selection, water composition and corrective additions.

The pH of a nutrient solution is a critical parameter that determines the speciation of elements across their various physicochemical forms (free ions, complexes, chelates, ion pairs, solid and gaseous phases and different oxidation levels) and is therefore critical for nutrient bioavailability (Rijck & Schrevens (1998)). The implemented pH estimation module provides a practical way to evaluate the expected pH of the formulated solution based on its composition and the applied acid or base correction. This allows users to anticipate whether the resulting pH will be within a desirable range for plant growth and to make informed decisions about formulation adjustments if necessary.

While both γ activity-correction models, Davies and Debye–Hückel, were evaluated against reference systems, the Davies model showed closer agreement with the phosphate-buffer pH data and with the KCl conductivity data at 25 °C. This was an important criterion for model selection because KCl solutions are standard reference solutions for EC meter calibration. In addition, Davies is more appropriate for the ionic-strength range of horticultural nutrient solutions than the limiting Debye–Hückel equation, since Davies is commonly considered applicable up to approximately $I \leq 0.5$ M, whereas the limiting Debye–Hückel equation is restricted to much more dilute solutions (Kalka, 2023).

Within the tested ranges, the implemented pH and EC models therefore provided useful chemically informed approximations rather than full geochemical simulations. EC estimation based on diffusion coefficients together with Davies activity correction agreed closely with the KCl reference values and with PHREEQC-based comparisons. Although the Debye–Hückel variant reproduced the EC values reported by Sonneveld & Straver (1994) slightly more closely in the recipe-level comparison, this was not interpreted as general model superiority because those values are not primary conductivity standards and may reflect recipe-specific empirical calibration.

The small positive EC bias of the mechanistic NutriCalc models in the recipe-level comparison was interpreted cautiously. Overestimation is chemically plausible because the present EC implementation does not fully model complexation, ion pairing, or other speciation effects that can reduce or modify the concentration and mobility of freely conducting ions. Davies was therefore retained as the default activity-correction model, while Debye–Hückel was kept as a comparative alternative.

These modules do not replace full geochemical simulation. Their value lies in practical workflow integration because users can evaluate not only whether a target vector can be approximated, but also what pH and EC consequences follow from the achieved formulation.

5.7 Relevance of the mass balance approach

"Judging from their publications and statements, it would seem that many investigators have grown crops in a solution chosen largely at random and that, where the results have been successful, they have recommended that solution as being specific for the particular crop concerned." (Steiner, 1961)

Optimal nutrient solutions can be formulated using the principle of mass balance (Langenfeld et al., 2022). Bugbee et al. (Bugbee & Noah James Langenfeld, 2024) reported a related set of mass balance based nutrient recipes for monocots, dicots and *Cannabis*. In contrast to empirically derived nutrient recipes, this approach seeks to relate nutrient supply directly to plant demand. Plant nutrient demand can be approximated from tissue nutrient composition in combination with plant water consumption, thereby linking absolute nutrient uptake to water uptake. Langenfeld et al. (2022) distinguish between active uptake (N, P, K, Mn), intermediate uptake (Mg, S, Fe, Cu, Zn, Mo, Ni, Cl) and passive uptake (Ca, B). Passively taken up nutrients such as Ca are closely related to transpired water and are therefore well suited to a mass balance approach. Since tissue analysis represents a destructive end point measurement, non linearities in active nutrient uptake kinetics, including transitions between high affinity and low affinity transport systems, are of limited relevance in this context. With both WUE and tissue analysis, the exact temporal dynamics of uptake do not need to be resolved explicitly, as the temporal dimension is subsumed into the final nutrient solution composition. Thus, although active and passive uptake differ mechanistically and kinetically, these differences converge in the relationship between cumulative water use and the plant tissue nutrient content observed at harvest.

On this basis, crop-specific target concentrations for the nutrient solution can be derived. Under ideal conditions, such a formulation would maintain the residual or recirculating solution in approximate balance over time, thereby minimising the accumulation or depletion of individual nutrients. The present framework is particularly relevant here because a biologically plausible demand vector is not automatically a chemically realisable nutrient solution. Even if tissue composition and water use define a meaningful elemental target, this still has to be translated into an ionic target and then into a feasible fertilizer composition. NutriCalc provides a transparent way to make this translation explicit.

A remaining limitation of tissue based mass balance approaches is the translation of total plant N into a chemically feasible nutrient solution target. Specifically, total N must be partitioned into nitrate and ammonium. To address this, the present work introduces a refinement of the mass balance approach in which total N is partitioned into NO_3–N and NH_4–N by imposing electroneutrality as an explicit formulation constraint. This step enables the derivation of nutrient solutions that are chemically plausible and electrochemically balanced. As an illustrative calculation, applying the electroneutrality-based partitioning logic to a total N concentration of 90 mg L^{-1}, derived from Langenfeld et al. (2022), gives 82 mg L^{-1} NO_3–N and 8 mg L^{-1} NH_4–N.

This does not yet constitute a full physiological model of nutrient uptake, but it is a meaningful methodological step. It bridges biological demand estimation and chemically constrained fertilizer realisation within one transparent computational framework.

5.8 Comparison with existing formulation approaches

Existing nutrient formulation practice spans fixed recipes, manual iterative balancing, spreadsheet tools and dedicated software. Classical approaches such as the universal algorithm of Sonneveld et al. (1999) are chemically informed and practically valuable, but they are not necessarily expressed as explicit constrained optimisation frameworks. Spreadsheet based calculators can be useful in practice, but often remain difficult to extend reproducibly and may obscure the underlying calculation logic. Other software such as HydroBuddy offers flexible nutrient formulation functionality, but NutriCalc contributes a distinct combination of explicit stoichiometric representation, weighted row normalised NNLS fitting, charge balance oriented evaluation and modular open source implementation in R.

The main contribution of NutriCalc is therefore not merely that it calculates fertilizer quantities, but that it formalises nutrient formulation as a transparent, reproducible and modifiable constrained optimisation problem. The R implementation supports programmable workflows and integration with other analyses, whereas the Shiny interface broadens accessibility for users without programming experience.

5.9 Limitations and scope

Despite its strengths, the framework has clear limitations. First, precipitation reactions are not explicitly modelled. They are only considered indirectly through BAK separation, which reduces practical incompatibility risk in concentrated stock solutions but does not predict precipitation. A target may therefore appear feasible in dissolved-ion terms while being unstable in concentrated stock solutions. Second, chemical speciation remains simplified. Major acid-base systems and fixed-charge chelates are represented, but a full complexation model is not implemented. Third, the pH and EC calculations remain approximate. Debye–Hückel and Davies activity-coefficient models are suitable for dilute to moderately concentrated systems, but they do not capture all non-ideal effects at higher ionic strengths.

Fourth, non-zero residuals may reflect limitations of the selected fertilizer matrix rather than general chemical infeasibility. Exact solvability is always conditional on the implemented fertilizer set, which necessarily represents a selected subset of possible compounds. Fifth, the optimisation currently targets nutrient fit rather than explicit minimisation of cost, number of salts or operational complexity. Although users can partly influence the returned solution through salt selection and weighting, these objectives are not yet formalised directly. Finally, the framework is a formulation and evaluation model rather than a crop-growth model. It can represent and assess nutrient solution chemistry, but it does not predict plant response, uptake kinetics or recirculating-system dynamics over time.

The validation performed in this thesis therefore concerns computational and physicochemical plausibility rather than biological crop response. Recipe reconstructions evaluate whether target nutrient vectors can be reproduced from the implemented fertilizer matrix. The phosphate-buffer and KCl comparisons evaluate selected pH and EC calculation routines under controlled reference conditions. The Sonneveld comparison evaluates agreement with a published formulation approach under the reconstructed assumptions. By contrast, the thesis does not validate plant-growth responses, nutrient-uptake dynamics, precipitation behaviour in concentrated stock solutions, long-term recirculation effects or crop-specific yield outcomes. These aspects require experimental or more detailed geochemical validation and define the boundary of the present computational-methodological contribution.

These limitations define the scope of the framework rather than invalidating its usefulness. NutriCalc should therefore be understood as a transparent and extensible formulation tool that captures major stoichiometric and physicochemical constraints, while leaving more advanced geochemical and biological dynamics to future development.

6 Conclusions and Outlook

The first research question addressed whether nutrient solution formulation can be represented as a stoichiometric optimisation problem. The results indicate that it can, provided that fertilizer compounds are chemically defined and encoded in a compound-by-nutrient matrix.

For the second research question, the crop-specific reconstruction results showed that published target vectors could generally be reproduced with low residual error. Exact matching, however, depended on whether the target composition was electroneutral and reachable from the selected fertilizer set.

The third research question concerned the implemented pH and EC routines. The reference-system checks supported these calculations within the tested conditions, particularly when Davies activity correction and ion-specific conductivity contributions were used.

The fourth research question compared NutriCalc with the Sonneveld formulation approach. The comparison showed that NutriCalc could reproduce the published formulation example while making the calculation pathway more explicit and reproducible.

The fifth research question addressed practical formulation constraints. The case studies demonstrated that, within the tested scenarios, the framework could handle source-water correction, acid adjustment, restricted fertilizer availability and mixed-water formulation.

Overall, the thesis supports the conclusion that nutrient solution formulation can be shifted from opaque recipe adjustment toward a transparent computational workflow. This conclusion applies to the implemented formulation domain and should not be extended to biological crop performance without additional experimental validation.

6.1 Scientific contribution

This thesis developed `NutriCalc`, an open-source computational R package with a Shiny interface for the formulation, analysis and optimisation of nutrient solutions in horticulture. The work formalised nutrient solution design as a constrained linear optimisation problem and implemented this formulation using non-negative least squares. In doing so, it linked stoichiometric conversion, target-based optimisation, source-water correction and chemically informed evaluation of pH and electrical conductivity within one transparent workflow.

The results showed that a broad range of published horticultural target solutions can be reproduced exactly or with only minor residual deviations. Where deviations remained, these were chemically interpretable and could generally be traced to fertilizer representation, electroneutrality constraints or chemically coupled nutrient sources. Beyond reproducing established recipes, the framework also demonstrated how nutrient formulation can be extended toward mass-balance-derived targets and charge-balance-based interpretation.

The scientific contribution is therefore not merely the implementation of a software tool, but the formalisation of nutrient solution formulation as a reproducible computational workflow that makes stoichiometric constraints, solver behaviour and physicochemical consequences explicit.

6.2 Practical relevance for horticulture

The practical relevance of the framework lies in its applicability to both research and horticultural practice. For researchers, NutriCalc provides a transparent tool for constructing and comparing nutrient treatments under explicitly defined chemical constraints. For practitioners, it offers a flexible way to formulate crop-specific nutrient solutions while accounting for local water quality, available fertilizers and physicochemical properties of the final solution.

By combining an R package with a Shiny-based graphical user interface, the framework is accessible both for programmable analytical workflows and for users without coding experience. In this sense, NutriCalc contributes not only methodologically but also operationally to digital nutrient management in horticulture.

6.3 Future extensions

Future development of NutriCalc could expand both the scientific scope of the framework and its practical applicability in horticultural research and nutrient solution management. One important direction is the expansion of the fertilizer database, including additional compounds used in horticultural practice, user-defined custom salts and further micronutrient sources such as Ni and Co. This would increase flexibility and allow the software to be adapted to different regional fertilizers and cultivation systems.

A second direction concerns the chemical model. Future versions could incorporate more detailed speciation, complexation and explicit precipitation modelling, particularly for concentrated stock solutions. This would strengthen the chemical realism of the framework and improve its applicability where solubility limits and ion interactions become practically relevant.

A third direction concerns the optimisation framework. At present, formulation is based on NNLS and focuses on matching target nutrient concentrations. Future versions could include additional objectives and constraints, such as fertilizer price, number of salts or operational simplicity. This would allow formulation to reflect not only chemical targets, but also economic and practical considerations.

A fourth direction concerns input flexibility, reporting and data integration. NutriCalc could be extended to accept target profiles in additional unit systems, import user-defined recipe datasets through formats such as CSV and generate structured PDF or CSV reports from the Shiny application. Such reports could document selected fertilizers, achieved nutrient concentrations, deviations from target values and predicted pH and EC.

A particularly important future extension is the integration of experimental data, especially for the mass-balance approach. This would allow tissue composition, water use and nutrient solution formulation to be linked more directly, supporting the development of more precisely balanced and crop-specific recipes for closed hydroponic systems.

7 Summary

This thesis addresses nutrient solution formulation in horticulture, where solution composition must be defined precisely under chemical and practical constraints. In practice, nutrient recipes are often derived from fixed formulations or manual trial-and-error adjustment. Such approaches are difficult to reproduce consistently across studies and prone to conversion errors, especially when water quality, fertilizer composition and different unit systems must be considered simultaneously.

To address this problem, the thesis developed `NutriCalc`, an open-source computational framework in R for the formulation, evaluation and optimisation of nutrient solutions. The central methodological contribution was the development of a reproducible computational pipeline that replaces ad hoc recipe formulation with mathematical optimisation. Nutrient solution design was formalised as a constrained linear optimisation problem and solved using non-negative least squares, allowing fertilizer combinations to be calculated transparently while ensuring physically meaningful, non-negative quantities.

The framework integrated stoichiometric conversion of fertilizer compounds, mapping to canonical nutrient targets, transformation between molar and mass-based units and correction for ions already present in the source water. In this way, NutriCalc established a consistent workflow linking fertilizer compounds, nutrient targets and practical concentration units. Beyond nutrient matching, it incorporated chemically informed evaluation of electroneutrality, pH and electrical conductivity, connecting optimisation results to physicochemical properties relevant for experimental interpretation and practical nutrient management. A contribution of the thesis was the implementation of the framework in two complementary forms: a programmable R package and a Shiny-based graphical user interface. This dual design supports both reproducible analytical workflows in research and interactive use in horticultural practice.

The results showed that a broad range of published horticultural target solutions could be reconstructed exactly or with only small residual deviations. Where exact fits were not possible, these deviations were generally chemically interpretable and could often be traced to identifiable constraints such as limited fertilizer choice, stoichiometric ion coupling across salts or electroneutrality requirements. The case studies further demonstrated that the framework can be applied to more complex scenarios involving alternative fertilizer availability, local water composition and mixed water sources. Overall, the thesis showed that nutrient solution design can be shifted from a recipe-based and partly opaque practice to a reproducible, optimisation-based and chemically explicit workflow, providing both a methodological framework for horticultural research and a practical basis for transparent nutrient management in soilless cultivation systems.

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Appendix

The appendix provides supplementary material for the evaluation and documentation of NutriCalc. It includes an additional comparison of reported and calculated electrical conductivity values for crop-specific nutrient-solution reconstructions, as well as the function documentation generated with `roxygen2` for the implemented R package. The complete R code is available in the GitHub repository at https://github.com/diolsa/nutricalc.

Table A.1: Comparison of reported EC values with calculated EC estimates for crop-specific recipe reconstructions under salts-only conditions. Calculations assume water = 0, KS 4.3 = 0 and CO2(aq) = 0. EC Sonneveld refers to the linear TCE-based estimate. Deviations are calculated as calculated EC minus reported EC.

Category	Recipe	EC reported (mS cm^{-1})	EC Sonneveld (mS cm^{-1})	EC Davies (mS cm^{-1})	EC DH (mS cm^{-1})	Deviation Sonneveld (mS cm^{-1})	Deviation Davies (mS cm^{-1})	Deviation DH (mS cm^{-1})
Olericulture	Endive in Recirculating Water	2.6	2.42	2.69	2.6	-0.18	+0.09	-0
	Eggplant in Rockwool	2.1	2.08	2.29	2.22	-0.02	+0.19	+0.12
	Eggplant in Rockwool (RDW)	1.7	1.62	1.8	1.76	-0.08	+0.1	+0.06
	Bean in Rockwool	1.7	1.67	1.82	1.77	-0.03	+0.12	+0.07
	Courgette in Rockwool	2.2	2.07	2.31	2.23	-0.13	+0.11	+0.03
	Cucumber in Rockwool	2.2	2.13	2.36	2.28	-0.07	+0.16	+0.08
	Cucumber in Rockwool (RDW)	1.7	1.65	1.81	1.76	-0.05	+0.11	+0.06
	Propagation vegetable plants in rockwool	2.4	2.39	2.6	2.49	-0.01	+0.2	+0.09
	Sweet Pepper in Rockwool	2.2	2.12	2.34	2.25	-0.08	+0.14	+0.05
	Sweet Pepper in Rockwool (RDW)	1.7	1.7	1.87	1.82	-0	+0.17	+0.12
	Lettuce in Recirculating Water	2.6	2.45	2.77	2.67	-0.15	+0.17	+0.07
	Tomato in Rockwool	2.3	2.34	2.59	2.48	+0.04	+0.29	+0.18
	Tomato in Rockwool (RDW)	1.6	1.62	1.79	1.75	+0.02	+0.19	+0.15
Floriculture	Alstroemeria in Rockwool	1.7	1.63	1.79	1.75	-0.07	+0.09	+0.05
	Alstroemeria in Rockwool (RDW)	1.1	1.25	1.33	1.31	+0.15	+0.23	+0.21
	Anemone in Rockwool	1.9	1.82	2	1.94	-0.08	+0.1	+0.04
	Carnation in Rockwool or Peat	1.8	1.79	1.98	1.92	-0.01	+0.18	+0.12
	Carnation in Rockwool (reuse drainage)	1.1	1.07	1.13	1.11	-0.03	+0.03	+0.01
	Anthurium andreanum in Rockwool or Peat	0.8	0.88	0.89	0.88	+0.08	+0.09	+0.08
	Aster in Rockwool or Peat	1.8	1.79	1.98	1.92	-0.01	+0.18	+0.12
	Bouvardia in Rockwool	1.9	1.89	2.06	2	-0.01	+0.16	+0.1
	Bouvardia in Rockwool (RDW)	1.2	1.25	1.31	1.29	+0.05	+0.11	+0.09
	Chrysanthemum in Recirculating Water	1.8	1.71	1.92	1.87	-0.09	+0.12	+0.07
	Cymbidium in phenol foam	0.8	0.9	0.9	0.89	+0.1	+0.1	+0.09
	Cymbidium in rockwool/urethane foam	0.8	0.89	0.9	0.88	+0.09	+0.1	+0.08
	Euphorbia fulgens in Rockwool	1.7	1.73	1.88	1.83	+0.03	+0.18	+0.13
	Freesia in Rockwool or Sand	2.1	1.98	2.22	2.15	-0.12	+0.12	+0.05
	Gerbera in Rockwool	1.7	1.63	1.79	1.74	-0.07	+0.09	+0.04
	Gerbera in Rockwool (RDW)	1.1	1.08	1.16	1.14	-0.02	+0.06	+0.04
	Gypsophila in Rockwool	2.2	2.16	2.33	2.24	-0.04	+0.13	+0.04
	Hippeastrum in Pumice	1.9	1.79	2	1.95	-0.11	+0.1	+0.05
	Pot Plants in Expanded Clay	1.6	1.54	1.68	1.64	-0.06	+0.08	+0.04
	Rose in Rockwool	1.6	1.6	1.73	1.69	+0	+0.13	+0.09
	Rose in Rockwool (RDW)	0.7	0.75	0.73	0.72	+0.05	+0.03	+0.02
	Statice in Rockwool	1.7	1.62	1.79	1.75	-0.08	+0.09	+0.05
Fruticulture	Strawberry in Peaty Substrates	1.7	1.67	1.83	1.78	-0.03	+0.13	+0.08
	Strawberry in Recirculating Water	1.5	1.48	1.6	1.56	-0.02	+0.1	+0.06
	Melon in Rockwool	2.2	2.17	2.4	2.31	-0.03	+0.2	+0.11

Package 'nutricalc'

May 3, 2026

Type Package

Title Nutrient Solution Formulation in Horticultural Sciences

Version 0.1.0

Description Provides tools for parsing chemical formulas, calculating molar masses, and optimizing fertilizer mixtures to meet specified nutrient targets using non-negative least squares. Designed to help agronomists and researchers formulate custom nutrient solutions efficiently.

License GPL-3

Encoding UTF-8

LazyData true

Depends R (>= 3.5)

Imports stringr,
nnls,
shiny,
bslib,
Ternary,
xml2

RoxygenNote 7.3.2

Contents

`assign_salts_bak` *Assign salts to A, B, and Micro tanks based on composition and solver result*

Description

This helper takes the result from `optimize_nutrients()` and the corresponding `nutrient_matrix` (compounds x nutrients) and returns a BAK-style grouping:

Usage

```
assign_salts_bak(
  res,
  nutrient_matrix,
  micro_nutrients = c("Fe", "Mn", "Zn", "B", "Cu", "Mo", "Si"),
  calcium_nutrient = "Ca",
  phosphorus_nutrient = "P",
  sulfur_nutrient = "S",
  balance_nutrients = c("NO3_N", "NH4_N", "P", "K", "Ca", "Mg", "S", "Na"),
  drop_zero = TRUE,
  tol = 0
)
```

Arguments

`res` A `nutrient_optimization_result` as returned by `optimize_nutrients()`.

`nutrient_matrix`
: Matrix [compound x nutrient] used in the optimization. Row names must match the salt names (formulas).

`micro_nutrients`
: Character vector of column names that count as micronutrients. Defaults to `c("Fe", "Mn", "Zn", "B", "Cu", "Mo")`.

`calcium_nutrient`
: Column name for calcium (default: `"Ca"`).

`phosphorus_nutrient`
: Column name for phosphorus (default: `"P"`).

`sulfur_nutrient`
: Column name for sulfur (default: `"S"`).

`balance_nutrients`
: Character vector of nutrient columns used to define the "load" when splitting AB salts between A and B. Defaults to `c("NO3_N","NH4_N","P","K","Ca","Mg","S","Na")`.

`drop_zero`	Logical, if `TRUE` (default), remove salts with effectively zero mmol/L from the returned `table` and `by_tank`. The `assignments` vector always includes all salts.
`tol`	Numeric tolerance for deciding what counts as "zero" mmol/L.

Details

- First, salts are classified purely by composition: * Micro = any micro nutrient present (e.g. Fe, Mn, Zn, B, Cu, Mo) * A = contains Ca (and no micro) * B = contains P or S (and no micro / Ca) * AB = none of the above (can go to A or B)

- Then the AB salts are assigned to A or B so that A and B have a similar overall "load" (sum of delivered mmol of selected nutrients).

Value

A list with:

`assignments`	Named character vector, salt -> tank ("A","B","Micro").
`table`	Data frame with columns `salt`, `tank`, `mmol_l` (filtered to non-zero if `drop_zero = TRUE`).
`by_tank`	Named list of data frames (A, B, Micro, possibly others), each subset of `table`.

`CANONICAL_NUTRIENTS` *Canonical nutrient keys for targets and UI*

Description

Defines the stable nutrient order used across recipes, UI inputs, and target vectors. This is intentionally independent from the full ‘nutrient_matrix‘, which may include additional columns for chelate ligands or other species.

Usage

```
CANONICAL_NUTRIENTS
```

Format

A character vector of nutrient keys.

compute_molar_mass *Compute molar mass of a chemical formula*

Description

Supports formulas like "NaCl", "Ca(NO3)2", "CuSO4·5H2O", "MgN2O6·6H2O"

Usage

```
compute_molar_mass(formula)
```

Arguments

`formula`	Chemical formula string

Value

Molar mass in g/mol

compute_percentage_matrix *Compute a matrix of mass percentages of nutrients in each fertilizer*

Description

Compute a matrix of mass percentages of nutrients in each fertilizer

Usage

```
compute_percentage_matrix(nutrient_matrix)
```

Arguments

`nutrient_matrix`	A [fertilizer x nutrient] matrix in mmol/mol

Value

A matrix of nutrient mass percentages per fertilizer

converte — *Convert amount of one compound to another using molar mass and element counts*

Description

This function computes the conversion factor between two chemical formulas or elements based on their molar masses and the count of shared atoms. It adjusts for stoichiometry where applicable (e.g., converting from "SO4" to "S").

Usage

```
converte(from, to, value = 1, element = NULL)
```

Arguments

`from`	A character string representing the source formula or element (e.g., "Ca", "NaCl", "SO4").
`to`	A character string representing the target formula or element.
`value`	A numeric value to convert (default is 1). Represents the quantity of ‘from‘.
`element`	Optional element symbol to anchor the stoichiometric conversion when multiple elements are shared between ‘from‘ and ‘to‘. If omitted and more than one element is shared, an error is raised to avoid ambiguous conversions.

Value

A numeric value representing the equivalent amount of ‘to‘, adjusted by molar mass and atom ratio.

Examples

```
converte("SO4", "S")        # Convert sulfate to elemental sulfur
converte("Na", "NaCl")      # Convert sodium to sodium chloride
converte("P2O5", "P")       # Convert phosphorus pentoxide to elemental phosphorus
converte("Ca", "CaCl2\u00b72H2O", 10)  # Convert 10 mmol of Ca to CaCl2\u00b72H2O
converte("(NH4)2HPO4", "NH4H2PO4", element = "P")  # Disambiguate using P as the anchor
```

convert_units — *Convert nutrient concentrations between mmol/L and mg/L*

Description

Converts numeric nutrient values (single element or named vector) between mmol/L and mg/L using molar masses in ‘element_molar_mass_df‘.

Usage

```
convert_units(x, element = NULL, to = c("mg/L", "mmol/L"))
```

Arguments

x	Numeric scalar or named numeric vector of nutrient values. If named, names should match those in ‘nutrient_element_map‘.
element	Optional element symbol (e.g. "N", "P") when ‘x‘ is a scalar.
to	Target unit: "mg/L" or "mmol/L".

Value

Numeric vector (same length/names as ‘x‘) in the target unit.

Examples

```
convert_units(c(NO3_N = 15, P = 1.5), to = "mg/L")
convert_units(c(NO3_N = 210, P = 46), to = "mmol/L")
```

---

drop_tiny *Zero-out tiny floating point values*

Description

Utility helper to clamp extremely small values to zero, avoiding noisy output.

Usage

```
drop_tiny(x, tol = 1e-09)
```

Arguments

x	Numeric vector or matrix.
tol	Values with absolute magnitude below this tolerance are set to zero.

Value

The input with near-zero entries replaced by zeros.

Examples

```
drop_tiny(c(1, 1e-10, -1e-12))
drop_tiny(matrix(c(0.5, 1e-12), nrow = 1))
```

ec_from_ph *Compute EC from achieved composition and pH speciation*

Description

Estimates conductivity using an ion-transport summation based on diffusion coefficients and activity coefficients.

Usage

```
ec_from_ph(
  achieved,
  ph_res,
  fe_state = c("Fe2+", "Fe3+"),
  t_C = 25,
  gamma = NULL,
  gamma_model = 1,
  A_DH_25 = 0.5085
)
```

Arguments

`achieved`	Named numeric vector (mmol/L) or data.frame with columns Nutrient and Achieved.
`ph_res`	Result list from ‘ph_from_achieved()‘ containing ‘species_mM‘.
`fe_state`	One of "Fe2+" or "Fe3+". When "Fe3+", all Fe2+ is treated as Fe3+ for EC purposes.
`t_C`	Temperature in degrees C.
`gamma`	Optional named numeric vector of activity coefficients (per ion).
`gamma_model`	Activity model selector: 1/"davies_25C" (Davies) or 2/"debye_huckel_25C" (Debye-Huckel). Default 1.
`A_DH_25`	Debye-Huckel A constant at 25 C (also used as A in Davies).

Details

Units: - c is in mmol/L (mM); numerically identical to mol/m^3, used directly as such. - D is in 10^-5 cm^2/s and converted to m^2/s by multiplying 1e-9. - Output: EC in mS/cm and uS/cm.

gamma_model: 1 = Davies (default) 2 = Debye-Hückel

Override precedence: model baseline -> ph_res$gamma_ions -> gamma argument (highest priority)

Value

A list with EC_mS_cm, EC_uS_cm, and per-ion contributions.

element_molar_mass_df *Atomic molar masses used for formula parsing*

Description

A lookup table of elemental (atomic) molar masses in grams per mole (g/mol). Used by molar-mass and unit-conversion helpers (e.g., when computing the molar mass of a chemical formula from its elemental composition).

Usage

```
data(element_molar_mass_df)
```

Format

A data frame with 23 rows and 2 columns:

Element Element symbol (character), e.g. `"C"`, `"O"`, `"Fe"`.

MolarMass_g_per_mol Atomic molar mass in g/mol (numeric).

fetch_bwb_mittelwert *Fetch BWB "In aller Tiefe" Mittelwerte by postal code*

Description

Retrieves Berliner Wasserbetriebe (BWB) analysis data for a given Berlin postal code and returns the Mittelwert values (first available entry per parameter) for selected parameters as mmol/L.

Usage

```
fetch_bwb_mittelwert(plz)
```

Arguments

`plz`	Character or numeric vector of length one with a five digit German postal code.

Value

A named numeric vector of length 18 in mmol/L with names ‘c("NO3_N","NH4_N","P","K","Ca","Mg","S","Na","Cl", "Fe","Mn","Zn","B","Cu","Mo","Si","KS4_3","KB8_2")‘, limited to the decimal precision provided by the source values. ‘KS4_3‘ is the acid capacity (alkalinity, KS 4.3); ‘KB8_2‘ is the base capacity (acidity, KS 8.2) and is treated as dissolved CO2(aq).

`ion_diffusion_25C` *Lookup table of ion diffusion coefficients at 25 C*

Description

Returns a data frame with ions, charge, and diffusion coefficients (D) at 25 C. Values are expressed as 10^-5 cm^2/s.

Usage

```
ion_diffusion_25C()
```

Value

A data.frame with columns: ion, z, D_1e5_cm2_s.

`nutrient_matrix` *Canonical fertilizer-to-nutrient stoichiometry matrix*

Description

A compound-by-nutrient matrix used by the nutrient solvers. Rows are fertilizer compounds (salts/acids/bases) and columns are nutrients. Each entry is the amount of nutrient delivered by 1 mmol of the compound (mmol nutrient per mmol compound).

Usage

```
data(nutrient_matrix)
```

Format

A numeric matrix with n rows (compounds) and 20 columns (nutrients). Row names are chemical formulas; column names are nutrient keys: ‘NO3_N‘, ‘NH4_N‘, ‘P‘, ‘K‘, ‘Ca‘, ‘Mg‘, ‘S‘, ‘Na‘, ‘Cl‘, ‘Fe‘, ‘Mn‘, ‘Zn‘, ‘B‘, ‘Cu‘, ‘Mo‘, ‘Si‘, ‘EDTA‘, ‘DTPA‘, ‘EDDHA‘, ‘HBED‘.

Details

The matrix entries are stoichiometric coefficients in mmol/mmol. For example, calcium nitrate (‘Ca(NO3)2‘) contributes 2 mmol ‘NO3_N‘ and 1 mmol ‘Ca‘ per 1 mmol of salt.

`optimize_nutrients` *Optimize Nutrient Solution Using NNLS*

Description

Solves for optimal fertilizer amounts to meet target nutrient concentrations. Expects a compound-by-nutrient matrix: rows = salts/compounds, columns = nutrients.

Usage

```
optimize_nutrients(nutrient_matrix, target, importance)
```

Arguments

`nutrient_matrix`	Matrix [compound x nutrient] with entries in mol of nutrient per mmol of compound (i.e., how many mmol of nutrient you get from 1 mmol of the compound).
`target`	Named numeric vector of target concentrations (mmol/L).
`importance`	Named numeric vector (-2 to +2) specifying nutrient importance (weights).

Value

An object of class 'nutrient_optimization_result' with elements:

- amounts - named numeric (mmol/L) of each compound (same order as rownames of input)
- achieved - named numeric (mmol/L) delivered per nutrient
- target - named numeric (mmol/L) target per nutrient
- abs_error - achieved - target (mmol/L)
- percent_error - 100 * abs_error / target (NA when target == 0)
- squared_error - sum(abs_error^2)
- rel_squared_error - sum((abs_error/target)^2, na.rm = TRUE)

`percentage_matrix` *Nutrient mass percentage matrix*

Description

Mass percentages of nutrients in each fertilizer derived from 'nutrient_matrix'.

Usage

```
data(percentage_matrix)
```

Format

A numeric matrix with rows as fertilizer formulas and columns as nutrients (NO3_N, NH4_N, P, K, Ca, Mg, S, Na, Cl, Fe, Mn, Zn, B, Cu, Mo, Si). Each value is the mass percentage of the nutrient in the fertilizer.

`ph_from_achieved` *Compute pH of a nutrient solution from achieved composition*

Description

Uses charge balance with acid/base speciation and activity corrections to estimate pH at 25 C. Assumes no carbonate alkalinity, no complexation, and no precipitation.

Usage

```
ph_from_achieved(
  achieved,
  temp_C = 25,
  phc_bracket = c(3, 9),
  tol = 1e-09,
  max_iter = 200,
  inner_max_iter = 50,
  debug = FALSE,
  chelate_z = c(EDTA = -4, DTPA = -5, EDDHA = -4, HBED = -4),
  gamma_model = 1
)
```

Arguments

`achieved`	Named numeric vector of mmol/L (e.g., solver result 'x$achieved'). If present, 'CO2_aq' is treated as dissolved free CO2(aq) (mmol/L), mapped from BWB "Basenkapazitaet KB 8,2" / KS 8.2 and used to fix carbonate speciation.
`temp_C`	Temperature in Celsius (only 25 C supported in this version).
`phc_bracket`	Numeric length-2 bracket for pHc = -log10([H+]) search.
`tol`	Root tolerance in meq/L.
`max_iter`	Maximum iterations for the outer bisection.
`inner_max_iter`	Maximum iterations for the ionic strength fixed point.
`debug`	Logical; when TRUE prints basic bracket diagnostics.
`chelate_z`	Named numeric vector of chelate charges to include as fixed anions. Values are net charges (negative) for ligand totals in mmol/L.
`gamma_model`	Integer 1 (Davies) or 2 (Debye–Hückel). Default 1.

Details

gamma_model: 1 = Davies (default) 2 = Debye–Hückel limiting law (DH)

Value

A list with pH, ionic strength, activity coefficients, species distribution, and charge balance diagnostics.

plot_ternary *Steiner-style ternary plot (Target vs Achieved) with constant-component drops*

Description

Steiner-style ternary plot (Target vs Achieved) with constant-component drops

Usage

```
plot_ternary(
  result,
  keys,
  title = "Ternary (ratios)",
  show_legend = TRUE,
  step_percent = 10,
  lab_cex = 0.95,
  axis_cex = 0.85,
  padding = 0.08,
  col_target = "black",
  col_achieved = "firebrick",
  pch_target = 16,
  pch_achieved = 17,
  cex_points = 1.25,
  show_link = FALSE,
  lty_link = 2,
  lwd_link = 1,
  legend_pos = "topright",
  show_axis_drops = TRUE,
  drop_col = "grey35",
  drop_lwd = 1,
  drop_lty = 1,
  show_ratio = TRUE,
  ratio_format = "percent",
  ratio_digits = 1
)
```

Arguments

`result` List with numeric named vectors ‘target‘ and ‘achieved‘.

`keys` Character(3) giving axis order, e.g. c("K","Ca","Mg").

`title` Plot title.

`show_legend` Logical, show legend (default TRUE).

`step_percent` Integer in [1, 50]; major grid step as percent. Default 10.

`lab_cex, axis_cex, padding`
Plot layout controls.

`col_target, col_achieved`
Colors.

`pch_target, pch_achieved`
Point symbols.

`cex_points` Point size.

`show_link` Logical, draw connector between Target and Achieved.

`lty_link, lwd_link` Style for connector if shown.

`legend_pos` Legend position.

`show_axis_drops` Logical, draw constant-component drops from Achieved to sides.

`drop_col, drop_lwd, drop_lty` Style for drop lines.

`show_ratio` Logical, include "(A, B; C)" values in legend (default TRUE).

`ratio_format` "percent" (default) or "proportion".

`ratio_digits` Digits for numeric formatting.

Value

Invisibly returns NULL (draws the plot).

---

print.bwb_mittelwert *Pretty-print BWB Mittelwerte*

Description

Pretty-print BWB Mittelwerte

Usage

```
## S3 method for class 'bwb_mittelwert'
print(x, ...)
```

Arguments

`x` A ‘bwb_mittelwert‘ numeric vector.

`...` Additional arguments passed to ‘print‘.

Value

The input ‘bwb_mittelwert‘ object, invisibly.

`print.nutrient_optimization_result`
Pretty-print the optimization result

Description

Shows per-liter amounts; if ‘vol > 1‘, also shows a dynamic column "g for <vol> L". Uses ‘compute_molar_mass(formula)‘ (g/mol) when available; if not found, prints without mass columns.

Usage

```
## S3 method for class 'nutrient_optimization_result'
print(x, vol = 1, ...)
```

Arguments

`x`	A ’nutrient_optimization_result’ object.
`vol`	Total solution volume in liters (must be > 0). If > 1, a "g for <vol> L" column is added.
`...`	Unused.

`recipes` *Nutrient solution recipes used by NutriCalc*

Description

A named list of recipe definitions. Each element is a list with:

name Display name

category Category such as "Olericulture", "Fruticulture", ...

notes Free text notes / citation

unit Either "mmol/L" or "mg/L"

salts TRUE (all salts) or a character vector of salt formulas

targets Named numeric vector of nutrient targets

Usage

```
recipes
```

Format

A named list of recipe lists.

`recipe_summary` *Create a summary table of nutrient concentrations in both units*

Description

Given a named vector of nutrient concentrations and its unit, returns a data frame showing each nutrient's value in mmol/L and mg/L.

Usage

```
recipe_summary(recipe, unit = c("mmol/L", "mg/L"))
```

Arguments

`recipe`	Named numeric vector of nutrient concentrations (e.g., NO3_N, P, K, ...).
`unit`	Current unit of the values: "mmol/L" or "mg/L".

Value

A data frame with columns ‘Nutrient‘, ‘Element‘, ‘mmol/L‘, ‘mg/L‘.

Examples

```
recipe_summary(c(NO3_N = 15, P = 1.5), unit = "mmol/L")
recipe_summary(c(NO3_N = 210, P = 46), unit = "mg/L")
```

`run_app` *Launch the NutriCalc Shiny App*

Description

Launch the NutriCalc Shiny App

Usage

```
run_app(...)
```

Arguments

`...`	Arguments passed to [shiny::runApp()].

salt_info — *Metadata for fertilizer compounds*

Description

Descriptive metadata for each compound in nutrient_matrix, including a human-readable category name and whether the compound is treated as an acid/base in the two-stage solver.

Usage

```
data(salt_info)
```

Format

A data frame with n rows and 3 columns:

salt Chemical formula key matching `rownames(nutrient_matrix)`.

category Human-readable compound name/category.

is_acid_base Logical flag; `TRUE` for acids/bases.

strip_negative_zero — *Remove negative zeros introduced by rounding*

Description

Ensures rounded values like -0.00 are displayed as 0.

Usage

```
strip_negative_zero(x)
```

Arguments

x	Numeric vector or matrix.

two_stage_optimize_nutrients
Two-stage nutrient optimization separating neutral salts from acids/bases

Description

Runs an initial NNLS optimization on neutral salts, then uses acids/bases to fill remaining deficiencies (when they can affect the off-target nutrients).

Usage

```
two_stage_optimize_nutrients(
  nutrient_matrix,
  target,
  importance,
  tol_abs = 0.01,
  tol_rel = 0.01
)
```

Arguments

`nutrient_matrix`	Matrix [compound x nutrient] with entries in mol of nutrient per mmol of compound.
`target`	Named numeric vector of target concentrations (mmol/L).
`importance`	Named numeric vector (-2 to +2) specifying nutrient importance (weights).
`tol_abs`	Absolute tolerance (mmol/L) below which residuals are ignored for acid/base correction.
`tol_rel`	Relative tolerance (fraction of target) below which residuals are ignored for acid/base correction.

Value

A 'nutrient_optimization_result' object.

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Declaration of independence

I certify that I have written this thesis independently and have not used any sources or aids other than those specified, that all statements taken verbatim or in spirit from other writings have been identified and that the thesis has not been used in the same or a similar form for other examinations and has not been submitted to any other examination authority.

I have clearly labelled all passages that are taken from other works in terms of wording or meaning (including translations) as borrowed material, stating the exact source (including the World Wide Web and other electronic data collections). This also applies to attached drawings, pictorial representations, sketches and the like.

In addition, I confirm that when using IT/AI-supported tools, I have listed these tools in full in the "Overview of tools used" below with their product name and version number, my source of supply (e.g. URL) and details of use, and that I have completed the checklist truthfully. Excluded from this are those IT/AI-supported writing tools that were classified by my responsible examination office as not subject to notification by the time I submitted my thesis ("whitelist"). In the preparation of this thesis, I have consistently acted independently and with control over the use of IT/AI-supported writing tools.

I am aware that in the event of violations of these principles, a procedure for attempted cheating or deception will be initiated in accordance with the subject-specific examination regulations and/or the interdisciplinary statutes governing admission, studies and examinations at Humboldt-Universität zu Berlin (ZSP-HU).

Berlin, 12.05.26

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Place, date, signature

Whitelist

The following programmes do not have to be listed or evaluated. These programmes can be used without further information:

- ✓ Microsoft Office, LaTeX, OpenOffice, iWork, Google Docs
- ✓ Google Scholar, ResearchGate, Web of Science
- ✓ Databases of the university library
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Overview of tools used

Completion instructions:

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Type of use	**Programme**	**If relevant: sections concerned** (Please indicate page numbers if not applicable to the entire document).
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Literature research		
Translations of texts		
Summarising sources		

Have content explained in a different way (e.g. constructs, methodological approaches, analyses)	ChatGPT	
Creating text sections that serve as a template	ChatGPT	
Revision of your own text elements (please indicate the number of pages of the paper)	ChatGPT	
Analysing data (e.g. writing codes, defining suitable evaluation methods, creating illustrations)	ChatGPT, Codex	R code, Shiny app, documentation
Spell checking, grammar and writing style	ChatGPT	
Visualisations for illustrative or decorative purposes		

Further explanations if necessary:

OpenAI ChatGPT was used with the latest available model at the respective time of use. It was used throughout the writing process for revision of my own text, grammar and style checking and improvement of scientific wording. OpenAI Codex and ChatGPT were used primarily for programming support, especially for creating and revising R functions, developing the Shiny app, troubleshooting code and preparing the documentation. The final content, scientific decisions and responsibility for the thesis remained with the author.